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AI System for Sinus CT Analysis

Graduation Project 2 prepared to obtain the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science

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SUPERVISOR CERTIFICATION

I certify that the preparation of this project, entitled

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was carried out under my supervision at the Faculty of Artificial Intelligence Engineering in partial fulfillment of the requirements for the degree of B.Sc. in the Department of Artificial Intelligence Engineering and Data Science.

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كلمة شكر وتقدير

الحمد لله رب العالمين، الذي وفقني وأعاني على إتمام هذا العمل، وما كان لهذا الجهد أن يكتمل لولا فضل الله أولاً، ثم دعم وتوجيه أشخاص كان لهم أثر كبير في هذه المسيرة العلمية.

أتقدم بأسمى آيات الشكر والتقدير إلى المهندس أنس عبد العزيز، الذي كان له الدور الأكبر في مرافقتي ودعمي خلال ثلاث سنوات من العمل والتعلم. فقد كان موجهاً صبوراً، وداعماً حقيقياً، لم يخل بعلمه ولا بوقته ولا بنصحه، وكان حاضراً في المراحل الصعبة قبل السهلة. لقد علمني أن النجاح لا يأتي دفعة واحدة، بل يبنى بالصبر، والمثابرة، والإصرار، وأن العمل العلمي يحتاج إلى دقة ومسؤولية وإيمان بالفكرة. فله مني كل الامتنان والتقدير على تحمّله، وتشجيعه، وثقته، وعلى كل ما قدمه من علم وخبرة وتوجيه كان له أثر واضح في إنجاز هذا المشروع.

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الملخص

يعالج هذا المشروع مشكلة تحليل صور التصوير الطبقي المحوري للجيوب الأنفية باستخدام تقنيات التعلم العميق. وتكمن أهمية هذه المسألة في أن صور الجيوب الأنفية تحتوي على بُنى تشريحية متداخلة، كما أن تقييمها يدوياً قد يكون مرهقاً ويتطلب خبرة اختصاصية. وتظهر تحديات إضافية بسبب محدودية البيانات، وعدم توازن الحالات المرضية، واحتمال ترافق أكثر من شذوذ تشريحي أو مرضي في الحالة الواحدة.

يقترح المشروع خط معالجة يعتمد أولاً على التقسيم التشريحي باستخدام نموذج SwinUNETR ، وذلك لتحديد مناطق مهمة مثل الجيب الفكي الأيمن، والجيب الفكي الأيسر، والتجويف الأنفي الأيمن، والتجويف الأنفي الأيسر، والبلعوم الأنفي. بعد ذلك، يتم استخراج مناطق اهتمام موجّهة تشريحياً، ثم تدريب نموذج تصنيف متعدد التسميات للكشف عن خمس علامات مرضية مختارة. وقد تم اعتماد نموذج ConvNeXt-Tiny كنموذج التصنيف النهائي بعد مقارنته مع نموذج Swin-T.

أظهرت نتائج التقسيم قدرة جيدة على تحديد البنى التشريحية، حيث بلغ متوسط معامل Dice النهائي 87.43%، وبلغ متوسط IoU مقدار 81.02% على مجموعة الاختبار. أما في مرحلة التصنيف، فقد حقق نموذج ConvNeXt-Tiny أداءً جيد وفقاً لمتوسط Macro F1 عبر الطيات حيث بلغ 78.49% ونتائج التنبؤ خارج الطية 69.28% OOF F1 . ومع ذلك، بقيت بعض التسميات، وخاصة شذوذ الجيب الفكي، أقل استقراراً بسبب انخفاض عدد العينات الإيجابية وترافقها المتكرر مع شذوذات أخرى.

كلمات مفتاحية: التصوير الطبقي المحوري للجيوب الأنفية، تحليل الصور الطبية، التقسيم التشريحي، SwinUNETR، مناطق الاهتمام، ConvNeXt-Tiny، التصنيف متعدد التسميات.

Abstract

This project addresses the problem of artificial-intelligence-assisted analysis of sinus computed tomography (CT) scans. Sinus CT interpretation is clinically important because it supports the assessment of sinonasal anatomy and pathology, but automatic analysis is challenging due to the three-dimensional nature of CT volumes, anatomical variability, limited annotated data, and strong class imbalance among abnormal findings.

The proposed solution follows a segmentation-assisted pipeline. First, a SwinUNETR model is trained to segment key anatomical structures in NasalSeg CT volumes. Second, anatomy-guided regions of interest (ROIs) are extracted from the CT scan and segmentation masks. Third, a multi-label classifier predicts five final abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. ConvNeXt-Tiny is selected as the final classifier after comparison with Swin-T.

The segmentation model achieved a mean Dice score of 0.8743 and mean IoU of 0.8102 on the test set. For classification, ConvNeXt-Tiny achieved a fold-level macro F1 score of 0.7849 and an out-of-fold macro F1 score of 0.6928, outperforming Swin-T in the final comparison. The results show that the proposed pipeline is promising, while also revealing important limitations caused by small positive sample counts, multi-label co-occurrence, and indirect anatomical targets such as turbinate abnormalities.

Keywords: Sinus CT, Medical Image Segmentation, SwinUNETR, Region of Interest, ConvNeXt-Tiny, Multi-Label Classification.

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Glossary of Terms and Symbols

Table 0.1: Glossary of terms and abbreviations

Term	English Equivalent / Description	Symbol / Abbreviation
Computed Tomography	Cross-sectional X-ray imaging modality used to visualize internal anatomy.	CT
Region of Interest	A selected image region used as model input for a specific label.	ROI
Dice Similarity Coefficient	Overlap metric used to evaluate segmentation masks.	Dice
Intersection over Union	Overlap metric between predicted and reference masks.	IoU
Convolutional Neural Network	Deep learning architecture based on convolutional operations.	CNN
Vision Transformer	Neural architecture that applies self-attention to image patches.	ViT
SwinUNETR	Transformer-based 3D U-shaped segmentation architecture.	SwinUNETR
ConvNeXt	Modern convolutional architecture inspired by transformer-era design choices.	ConvNeXt
Out-of-Fold Prediction	Prediction produced for each sample by a model that did not train on it.	OOF
Focal Loss	Classification loss that reduces the contribution of easy examples.	Focal Loss
Average Precision	Area under the precision-recall curve, useful for imbalanced classification.	AP
Exact Match Accuracy	Multi-label metric requiring all labels of a case to be predicted correctly.	Subset Accuracy

Chapter One: Introduction

1.1 Medical Context and Sinus CT Interpretation

Computed tomography (CT) is one of the main imaging modalities used to evaluate the paranasal sinuses and nasal cavity. It provides detailed visualization of air-filled sinus cavities, bony boundaries, nasal passages, and soft-tissue changes. This makes sinus CT important for assessing sinonasal disease, anatomical variation, and preoperative planning in otolaryngology.

The interpretation of sinus CT is not performed as a single-image decision. A CT scan is a three-dimensional volume composed of multiple slices. In clinical reading, the radiologist reviews the scan through axial, coronal, and sagittal views, moving across slices and relating visible findings to their anatomical locations. Therefore, the interpretation process is naturally region-based: the reader localizes structures such as the maxillary sinuses, nasal cavities, nasal septum, and turbinate regions before describing possible abnormalities.

This medical workflow is important for artificial-intelligence-based analysis. A direct classifier that receives the full CT volume without localization may learn irrelevant visual patterns or focus on dominant structures while ignoring small or region-specific abnormalities. In contrast, a segmentation-guided system can first localize the relevant sinonasal anatomy, then use this localization to extract regions of interest for classification.

This project follows that clinical logic. The proposed system first performs anatomical segmentation using SwinUNETR, then extracts anatomy-guided regions of interest, and finally predicts selected abnormality labels using a ConvNeXt-Tiny classification model. The purpose is not to replace radiological interpretation, but to develop a research prototype that connects anatomical localization with multi-label abnormality prediction in sinus CT analysis.

1.2 Scientific Problem of the Research Project

The scientific problem addressed in this project is the automatic analysis of sinus CT scans for selected anatomical and pathological findings. The problem is challenging because sinus CT interpretation is both volumetric and anatomy-dependent. The same abnormality cannot be

evaluated reliably without knowing the anatomical region in which it appears. For example, a right maxillary sinus abnormality should be assessed from the right maxillary sinus region, while inferior turbinate hypertrophy is related to the nasal cavity region.

The clinical description of sinus CT findings is also not limited to one exclusive class. A single CT case may contain multiple co-existing findings, such as nasal septum abnormality, turbinate hypertrophy, and maxillary sinus abnormality. Therefore, the classification problem is naturally multi-label at the case level. A model must be able to predict more than one positive finding for the same patient rather than forcing the case into one dominant class.

The problem also includes important data-related limitations. The available dataset contains 130 matched CT cases, which is limited for deep learning. The final abnormality labels are imbalanced: nasal septum abnormality and inferior turbinate hypertrophy have more positive cases, while right and left maxillary abnormalities have only 22 and 23 positive cases. This imbalance makes learning and evaluating rare labels less stable.

Another challenge is the mismatch between available segmentation masks and some diagnostic targets. The NasalSeg dataset provides masks for the right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. However, it does not provide direct masks for the nasal septum or inferior turbinates. As a result, the system must use contextual anatomical regions for these targets instead of direct target masks.

The research problem is therefore not only to classify CT images, but to design an AI pipeline that respects the anatomical structure of the task. The system must localize relevant sinonasal anatomy, extract meaningful regions of interest, handle limited and imbalanced labels, and produce multi-label predictions that reflect the possibility of co-existing abnormalities.

1.3 Research Objectives

The main objective of this project is to develop a segmentation-assisted AI system for sinus CT analysis that follows a region-based medical workflow: first localizing relevant sinonasal anatomy, then extracting anatomy-guided regions of interest, and finally predicting selected abnormality labels.

The secondary objectives are:

- Train a 3D SwinUNETR segmentation model to localize the available NasalSeg anatomical structures: right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx.
- Convert the clinical annotation workbook into a model-ready multi-label format for five selected abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy.
- Design anatomy-guided ROI extraction rules that connect each abnormality label to the most relevant available segmentation region.
- Handle missing direct masks for the nasal septum and inferior turbinates using medically justified contextual ROIs.
- Evaluate ConvNeXt-Tiny and Swin-T as classification backbones under the same 5-fold out-of-fold evaluation setting.
- Select the best-performing final classifier and analyze its strengths, weaknesses, and limitations, especially under class imbalance and label co-occurrence.

1.4 Research Contribution

The main contribution of this project is the development of a segmentation-assisted multi-label classification workflow for sinus CT analysis. Instead of classifying the full CT volume without localization, the proposed system first segments relevant sinonasal structures and then uses these masks to extract anatomy-guided ROIs. This links each abnormality prediction to a clinically meaningful region and makes the pipeline more interpretable than a blind full-volume classifier.

A second contribution is the final ROI design. Because NasalSeg does not directly provide septum or turbinate masks, the project explores contextual regions based on the available nasal cavity masks. The final septum ROI is defined using both nasal cavities with padding, representing the idea that the septum lies between the right and left nasal cavity regions. The inferior turbinate labels are modeled using the ipsilateral nasal cavity ROIs.

A third contribution is the comparative evaluation of ConvNeXt-Tiny and Swin-T for this specific multi-label sinus CT task. ConvNeXt-Tiny is selected as the final model because it achieves better fold-level and out-of-fold performance than Swin-T. The report also discusses why some labels, especially maxillary abnormalities, remain unstable due to low positive counts and label co-occurrence.

The contribution is therefore not only the use of ConvNeXt-Tiny as a classifier, but the complete medical-AI workflow that connects sinus CT segmentation, ROI generation, multi-label abnormality prediction, and careful evaluation under limited-data conditions.

1.5 Target Beneficiaries

- Radiologists and ENT physicians who interpret sinus CT scans.
- Hospitals and imaging centers interested in AI-assisted triage or decision support.
- Researchers working on medical image segmentation and classification.
- Students and engineers studying applied deep learning in medical imaging.
- Developers who may extend the system into a clinical review or educational tool.

1.6 Report Organization

Chapter Two presents the theoretical background needed to understand the project, including CT imaging, sinus anatomy, segmentation, classification, CNNs, transformers, and evaluation metrics. Chapter Three reviews related work and identifies the research gap. Chapter Four explains the research methodology, dataset, preprocessing, segmentation model, ROI extraction, and final classification models. Chapter Five presents the experimental results and discusses the segmentation and classification outcomes. Chapter Six concludes the report and proposes future work.

Chapter Two: Theoretical Background

2.1 Computed Tomography and Sinus Imaging

Computed tomography constructs cross-sectional images from X-ray measurements acquired around the patient. In sinus imaging, CT is especially valuable because it shows the bony boundaries of the paranasal sinuses and nasal cavity. The intensity values in CT images are commonly represented in Hounsfield Units (HU), where air has very low values and bone has high values. This contrast is useful for distinguishing air-filled cavities from surrounding structures.

Sinus CT volumes are usually analyzed in axial, coronal, and sagittal planes. A model trained on CT data must therefore learn three-dimensional spatial relationships, or approximate them through multi-view two-dimensional representations. In this project, segmentation is performed as a 3D task, while classification uses 2.5D ROI images derived from each anatomical crop.

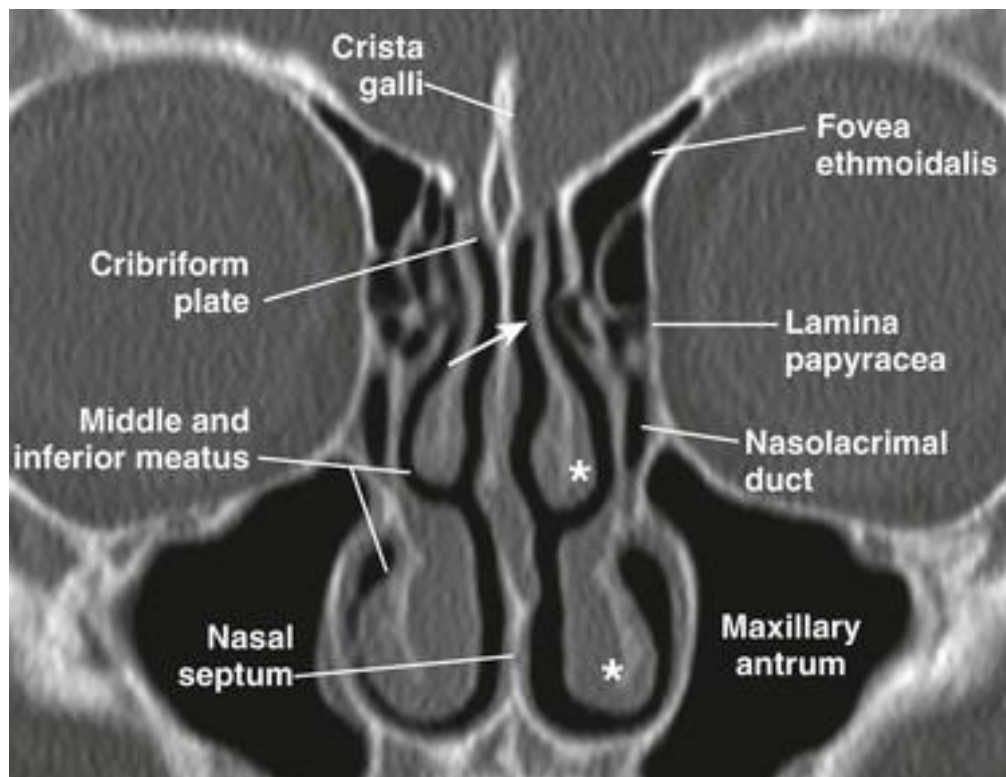


Figure 1: Main sinonasal anatomical regions in CT imaging

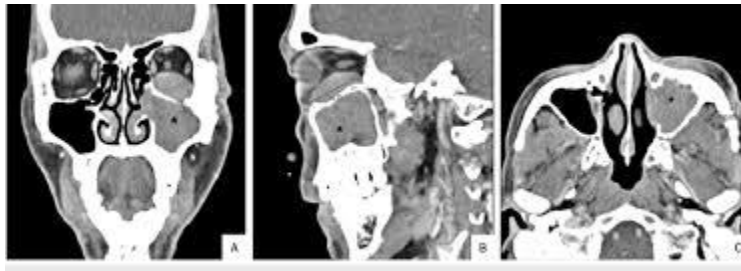


Figure 2: Example CT views of the sinonasal area, coronal, sagittal, axial

In this project, CT imaging is important not only as a diagnostic modality but also as the input source for both segmentation and classification. The air-filled sinus cavities, bony boundaries, and soft-tissue changes provide different intensity patterns that can be learned by the model. This is why CT windowing and multi-plane analysis are important for the proposed pipeline. The segmentation stage uses the 3D CT volume to localize anatomy, while the classification stage uses axial, coronal, and sagittal ROI views to preserve clinically useful context around each selected sinus region.

2.2 Sinonasal Anatomy Relevant to the Project

The paranasal sinuses are air-filled cavities surrounding the nasal cavity. In this project, the most important sinus structures are the right and left maxillary sinuses because they are directly related to the right and left maxillary abnormality labels. These regions are suitable for ROI-based analysis because the NasalSeg dataset provides separate segmentation masks for both maxillary sinuses.

The nasal cavity is divided into right and left sides by the nasal septum. Septal abnormality is clinically important because deviation or structural abnormality of the septum can affect nasal airflow and may coexist with other sinonasal findings. However, the dataset used in this project does not provide a direct nasal septum mask. Therefore, the septum label is handled using a contextual ROI derived from both nasal cavities, since the septum lies between them.

The inferior turbinates are anatomical structures located inside the nasal cavity and are involved in airflow regulation. Inferior turbinate hypertrophy can narrow the nasal airway and may appear together with septal or sinus abnormalities. Since direct turbinate masks are not available, the right and left inferior turbinate hypertrophy labels are modeled using the corresponding right and left nasal cavity ROIs.

Therefore, the anatomical background directly affects the design of the AI pipeline. Maxillary abnormality uses direct maxillary sinus ROIs, nasal septum abnormality uses a bilateral nasal cavity context ROI, and inferior turbinate hypertrophy uses ipsilateral nasal cavity ROIs. This connection between anatomy and ROI design is one of the main reasons the proposed system is more interpretable than a full-volume classifier.

2.3 Medical Image Segmentation

Medical image segmentation is the task of assigning a class label to each pixel or voxel. In 3D CT segmentation, each voxel in the volume is assigned to an anatomical class such as background, right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, or nasopharynx. Segmentation is useful because it localizes structures and enables measurement, visualization, and region-specific analysis.

In this project, segmentation is not the final clinical output. Instead, it is an intermediate stage used to guide ROI extraction. This design allows the classification model to focus on relevant anatomical regions rather than the entire CT volume.

$$Dice = 2 |P \cap G| / (|P| + |G|)$$

Equation 1: Dice Similarity Coefficient

$$IoU = |P \cap G| / |P \cup G|$$

Equation 2 : Intersection over Union

2.4 Deep Learning for CT Analysis

Deep learning models learn hierarchical representations from data. Early layers usually capture simple intensity and edge patterns, while deeper layers capture more complex anatomical or pathological features. For CT analysis, the model must be robust to anatomical variation, acquisition differences, and class imbalance.

There are two broad approaches to CT learning. A 3D model processes the full volume or volume crops and can preserve volumetric context, but it requires more memory and more data. A 2D or 2.5D model processes slices or multi-plane views, reducing memory requirements and enabling the use of pretrained natural-image backbones. This project uses 3D modeling for segmentation and 2.5D transfer learning for classification.

2.5 Convolutional Neural Networks and ConvNeXt

Convolutional neural networks are well suited for medical images because they exploit local spatial structure. ConvNeXt is a modern convolutional architecture that updates classical CNN design choices while preserving the simplicity and locality of convolutional operations. In this project, ConvNeXt-Tiny is used as the final classification backbone because it showed the best balance between performance and stability on the selected multi-label task.

The selection of ConvNeXt-Tiny is supported by two ideas. First, ConvNeXt preserves convolutional inductive bias, which is often helpful in limited-data medical imaging. Second, its design includes transformer-era improvements while avoiding some of the data demands of pure transformer backbones. This makes it a reasonable candidate for ROI-based 2.5D classification, where the input already focuses on a clinically meaningful region.

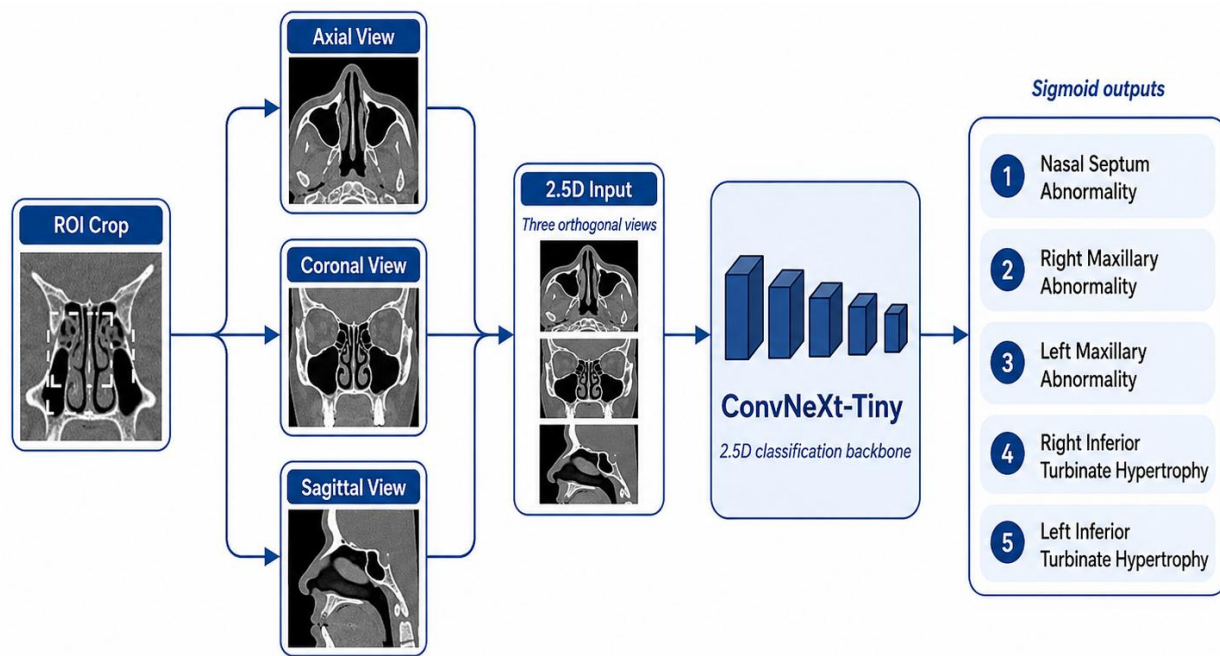


Figure 3: Simplified ConvNeXt-Tiny classification concept

2.6 Vision Transformers, Swin Transformer, and SwinUNETR

Vision transformers process images through patch representations and self-attention. Standard transformers can be computationally expensive for high-resolution medical images. Swin

Transformer addresses this issue by using hierarchical shifted windows, which makes attention more efficient while still capturing contextual information. SwinUNETR combines a Swin Transformer encoder with a U-shaped segmentation decoder, making it suitable for 3D medical segmentation tasks.

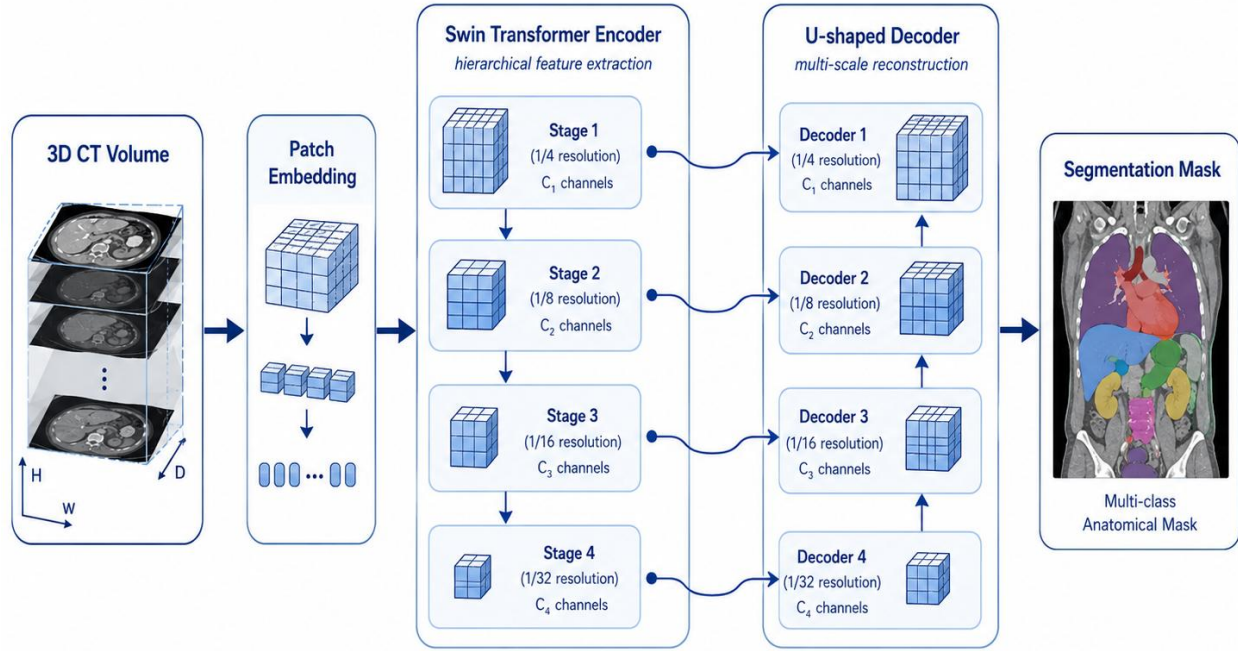


Figure 4: Simplified SwinUNETR architecture overview

2.7 Binary and Multi-Label Classification

Binary classification is a classification setting in which the model answers a yes-or-no question. In medical-image analysis, this usually corresponds to deciding whether a specific finding is present or absent. For example, a model may be asked whether right maxillary abnormality is present in a right maxillary sinus ROI.

Multi-label classification extends this idea to the case level. In multi-label classification, one case may have more than one positive label at the same time. This differs from multi-class classification, where each sample belongs to exactly one mutually exclusive class. In sinus CT analysis, this distinction is important because one patient may have nasal septum abnormality, maxillary sinus abnormality, and inferior turbinate hypertrophy in the same scan.

Therefore, the final diagnostic formulation in this project is multi-label at the CT-case level. Each abnormality label is treated as an independent binary target, and the final case output is

represented as a vector of five binary decisions:

$$\hat{y}_{\text{case}} = [\hat{y}_{\text{septum}}, \hat{y}_{\text{R-maxillary}}, \hat{y}_{\text{L-maxillary}}, \hat{y}_{\text{R-turbinate}}, \hat{y}_{\text{L-turbinate}}]$$

where each element indicates the predicted absence or presence of the corresponding abnormality. A normal case with none of the selected abnormalities is represented as an all-zero vector.

Sigmoid activation is suitable for this formulation because each output is interpreted independently as the probability of one abnormality being present. In contrast, a single softmax over the five abnormality labels would be inappropriate because it would force the model to choose only one abnormality. A two-class softmax can be used for a single binary task, but sigmoid is simpler and more consistent with the final case-level multi-label formulation.

$$\hat{y}_i = 1 \text{ if } \text{sigmoid}(z_i) \geq \tau_i, \text{ otherwise } 0$$

Equation 3: Binary classification decision from sigmoid probability

2.8 Class Imbalance, Positive Weighting, and Focal Loss

Class imbalance occurs when some labels have many positive examples while others have very few. This is an important issue in the proposed sinus CT classification task because the positive counts are not balanced across the five abnormality labels. In particular, the maxillary abnormality labels have fewer positive cases than the nasal septum and inferior turbinate labels. As a result, a model may learn frequent labels more easily while producing unstable predictions for rare labels.

In medical AI, this problem is important because rare positive findings should not be ignored. A model may achieve a high apparent accuracy by predicting mostly negative labels, but such behavior would be weak for clinical decision support. Therefore, evaluation must include recall, precision, F1-score, and per-label analysis rather than relying only on accuracy.

Positive class weighting is one way to reduce the bias toward negative predictions. It increases the training penalty when the model misses positive examples, especially for underrepresented labels. However, weighting cannot create new clinical diversity. If a label has very few positive cases, weighting can help the model pay more attention to that label, but it cannot fully replace the need for more annotated positive samples.

Focal loss was also used because it gives more attention to difficult or misclassified examples and reduces the relative influence of easy examples. In this project, focal loss was used with

gamma equal to 1. This setting was selected to address imbalance without making training overly aggressive for rare labels. The purpose of combining focal loss and positive weighting was to improve balanced behavior across labels, not simply to increase overall accuracy.

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

Equation 4: Binary focal loss concept

2.9 Evaluation Metrics

2.9.1 Segmentation Metrics

The segmentation model was evaluated with spatial overlap metrics and runtime measures. Dice and IoU measure agreement between predicted and reference masks. Precision measures the proportion of predicted foreground voxels that are correct, while recall or sensitivity measures the proportion of reference voxels captured by the model. Inference time and throughput describe the practical speed of the trained model during test-time prediction.

$$Precision = TP / (TP + FP)$$

Equation 5: Precision

$$Recall = TP / (TP + FN)$$

Equation 6: Recall / Sensitivity

$$Specificity = TN / (TN + FP)$$

Equation 7: Specificity

2.9.2 Classification Metrics

The classification task was evaluated using complementary metrics because the final task is imbalanced and multi-label. Accuracy alone is not sufficient, since a model can obtain high accuracy by correctly predicting frequent negative labels while still missing rare positive findings.

Precision measures how many predicted positive labels are actually correct. Recall measures how many true positive labels are detected by the model. Specificity measures how well the model rejects negative cases. Balanced accuracy averages sensitivity and specificity, making it more informative when positive and negative cases are imbalanced.

F1-score is the harmonic mean of precision and recall. Macro F1 is calculated by computing the F1-score separately for each abnormality label and then averaging the five label-wise F1 values. This gives each abnormality equal importance regardless of how many positive cases it has. For

this reason, macro F1 is one of the most important metrics in this project.

The report also uses out-of-fold evaluation. In 5-fold evaluation, each case is predicted once by a model that did not train on it. The out-of-fold predictions from all folds are then combined into one full prediction table, and the final OOF metrics are calculated from this combined table. This is more informative than relying only on a single train-test split, especially with only 130 matched CT cases.

Overall label accuracy measures correctness across all individual label decisions. Exact match accuracy is stricter: it counts a CT case as correct only if all five labels are predicted correctly at the same time. Therefore, exact match accuracy is expected to be lower than overall label accuracy in multi-label classification.

ROC and precision-recall curves can also be used as supporting analyses. ROC curves show the ranking behavior of the model across thresholds, while precision-recall curves are especially useful for imbalanced labels because they focus on positive-class detection. However, the final model selection in this project relies mainly on OOF macro F1, per-label F1, and threshold-tuned multi-label metrics.

$$F1 = 2 \times Precision \times Recall / (Precision + Recall)$$

Equation 8: F1-score

$$Balanced\ Accuracy = (Sensitivity + Specificity) / 2$$

Equation 9: Balanced accuracy

$$Exact\ Match = \text{number of cases with all labels correct} / \text{total number of cases}$$

Equation 10: Exact match accuracy

2.10 Chapter Summary

This chapter presented the theoretical background needed for the proposed sinus CT analysis system. It explained CT imaging, relevant sinonasal anatomy, medical image segmentation, 3D and 2.5D deep learning, ConvNeXt-Tiny classification, SwinUNETR segmentation, binary and multi-label classification, class imbalance, focal loss, and evaluation metrics. The chapter also clarified why the proposed task is multi-label at the CT-case level and why sigmoid-based binary decisions are suitable for the selected abnormality labels. These concepts support the

methodology described in the next chapter, where segmentation-guided ROIs are used for anatomy-aware abnormality prediction.

Chapter Three: Literature Review

3.1 Introduction

This chapter reviews previous studies related to sinus CT segmentation, paranasal anomaly classification, medical-image backbones, and 2.5D transfer learning. Similar to the organization used in the ChromoSep-AI literature review, the chapter discusses the most relevant studies separately rather than reducing them to a single table. This structure makes it clearer how each work influenced the design of the proposed segmentation-assisted sinus CT analysis system.

The reviewed literature shows three important directions. The first direction focuses on anatomical segmentation of sinus and nasal cavity structures. The second direction focuses on narrower abnormality-classification tasks, especially maxillary sinus anomaly detection. The third direction includes modern classification backbones and 2.5D medical-imaging strategies that can be adapted for limited-data CT classification. The gap addressed in this project is the connection between these directions: using anatomical segmentation to guide ROI-based multi-label abnormality prediction.

3.2 Previous Studies in Sinus CT Segmentation

Previous sinus CT segmentation studies mainly focused on localizing anatomical structures such as the maxillary sinuses, nasal cavity, nasopharynx, and other paranasal sinus regions. These works are important for this project because anatomical localization is the first stage of the proposed pipeline. Instead of using segmentation as the final clinical output, this project uses segmentation to guide ROI extraction before abnormality classification. The reviewed segmentation papers therefore provide the technical foundation for using 3D models such as SwinUNETR and U-Net variants in sinonasal CT analysis.

3.3 NasalSeg Dataset and Benchmark

Zhang et al. introduced NasalSeg, an open-access dataset for nasal cavity and paranasal sinus segmentation from 3D CT images [1]. The dataset contains 130 CT scans with voxel-wise annotation of five structures: right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. This work is the most directly related foundation for the present project because the proposed system uses the same anatomical structures to guide ROI

extraction. However, NasalSeg does not directly annotate the nasal septum or inferior turbinates, which creates an important limitation for downstream abnormality classification.

3.4 Pani et al. 3D-NASE Ensemble for NasalSeg Segmentation

Pani et al. proposed 3D-NASE, an ensemble-based framework for nasal cavity and paranasal sinus segmentation on the NasalSeg dataset [19]. The method is relevant to this project because it evaluates segmentation performance on the same anatomical segmentation problem used as the foundation for ROI extraction. Their reported results showed strong segmentation performance, with Dice and mIoU values that demonstrate the feasibility of accurate anatomical localization in NasalSeg-style CT data.

The importance of this study for the present project is not that it solves the same final task, but that it strengthens the evidence that reliable segmentation can be achieved for sinonasal CT structures. However, similar to the NasalSeg benchmark itself, the work remains focused on anatomical segmentation and does not address downstream multi-label abnormality prediction. Therefore, it supports the first stage of this project but does not replace the need for the proposed segmentation-guided classification pipeline.

3.5 SwinUNETR for 3D Medical Image Segmentation

Hatamizadeh et al. proposed SwinUNETR, a transformer-based encoder-decoder architecture for 3D medical image segmentation [3]. The model uses a Swin Transformer encoder to capture hierarchical volumetric context and a U-shaped decoder to reconstruct dense segmentation masks. In their medical segmentation benchmarks, transformer-based 3D modeling showed strong Dice performance and motivated its use for volumetric anatomical localization. In this project, SwinUNETR was selected for the segmentation stage because sinus CT volumes require spatial context across slices.

3.6 Whangbo et al. Paranasal Sinus Segmentation

Whangbo et al. compared several 3D U-Net variants for multi-class paranasal sinus segmentation in sinusitis CT images [6]. The normal 3D U-Net achieved the best reported F1 score, reaching 84.29% on the normal test set and 79.32% on the abnormal test set. This study supports the use of 3D CNN-based segmentation for sinus anatomy, especially when sinusitis changes the appearance of the target regions. Its limitation for the present project is that it focuses on

anatomical segmentation rather than multi-label abnormality prediction.

3.7 Ozturk et al. Maxillary Sinus Segmentation

Ozturk et al. developed a U-Net-based method for automatic maxillary sinus segmentation on cone-beam CT images [9]. The study is relevant because maxillary sinus abnormality is one of the final target groups in this project. It demonstrates that deep learning can localize maxillary sinus boundaries effectively, which is important for region-specific analysis. However, the work is CBCT-focused and does not address a full sinus CT pipeline with septum, turbinate, and bilateral multi-label abnormalities.

3.8 Song et al. CNN, Transformer, and Hybrid Segmentation

Comparison

Song et al. compared CNN, vision-transformer, and hybrid networks for paranasal sinus segmentation in CT images [10]. Their results showed that Swin UNETR achieved the strongest overall segmentation performance among the tested models, with a mean Jaccard Index of 0.719, Dice Similarity Coefficient of 0.830, recall of 0.758, and HD95 of 10.529. This study is important because it directly compares architecture families in the sinus domain. It also supports the idea that hybrid 3D architectures are strong candidates for sinus segmentation, although it remains focused on segmentation rather than diagnosis.

3.9 MedNeXt and ConvNeXt-Style Medical Networks

Roy et al. introduced MedNeXt, a ConvNeXt-inspired architecture for medical image segmentation [12]. The model adapts large-kernel convolutional design to 3D encoder-decoder segmentation and shows that modern CNN principles can remain competitive in medical imaging. This supports the broader idea that convolutional models should not be dismissed in favor of transformers only. For the present project, this is conceptually important because ConvNeXt-Tiny achieved the best classification behavior on the limited ROI-based dataset.

3.10 Previous Studies in Sinus and Paranasal Abnormality

Classification

Previous classification studies in sinus and paranasal imaging have commonly focused on narrower diagnostic targets, especially maxillary sinus anomaly detection. These studies show that deep learning can support abnormality recognition in CT or CBCT images, but most of them

do not address a five-label multi-label setting. In contrast, the current project predicts septum abnormality, bilateral maxillary abnormality, and bilateral inferior turbinate hypertrophy from the same CT case. This makes the classification problem more difficult because labels may co-occur and some target structures are not directly segmented.

3.11 Bhattacharya et al. Self-Supervised Maxillary Anomaly Classification

Bhattacharya et al. proposed a self-supervised learning approach for classifying paranasal anomalies in the maxillary sinus [7]. The method used a 3D convolutional autoencoder and a localization-oriented self-supervised task before fine-tuning for normal-versus-anomalous maxillary sinus classification. When trained with only 10% of the annotated dataset, the proposed method achieved an AUPRC of 0.79, outperforming BYOL, SimSiam, SimCLR, and SparK baselines. This is relevant to the present project because it addresses limited annotation, but it remains narrower because it focuses on maxillary sinus anomalies rather than five-label sinus CT classification.

3.12 Supervised Contrastive Learning for Maxillary Anomalies

Bhattacharya et al. also studied supervised contrastive learning for classifying paranasal anomalies in the maxillary sinus [8]. Their 3D CNN trained with supervised contrastive learning and cross-entropy achieved an AUROC of 0.85, compared with 0.66 for a 3D CNN trained with cross-entropy alone. This result shows that representation learning can improve classification in limited-data paranasal anomaly settings. However, the task was still focused on maxillary sinus anomaly classification and did not solve the broader problem of multi-label sinus CT abnormality prediction.

3.13 Classification Backbones and Transfer Learning Studies

The classification stage of this project is also connected to broader backbone and transfer-learning studies. Modern CNNs such as ConvNeXt and hierarchical transformers such as Swin Transformer provide strong pretrained feature extractors that can be adapted to medical images. Because the available dataset is small, transfer learning is more practical than training a large classifier from scratch. The reviewed backbone studies justify the final comparison between

ConvNeXt-Tiny and Swin-T and support the use of 2.5D ROI images as inputs to pretrained 2D models.

3.14 ConvNeXt

Liu et al. introduced ConvNeXt as a modernized convolutional architecture that adopts several transformer-era design choices while preserving the inductive bias of CNNs [4]. The ConvNeXt family achieved competitive large-scale vision performance, with the overall family reaching 87.8% ImageNet top-1 accuracy in the original work. This is relevant to the present project because the final classifier uses localized 2.5D ROI images and limited medical data, where convolutional locality can be beneficial. ConvNeXt-Tiny was therefore selected as the final backbone after outperforming the transformer comparison in the project experiments.

3.15 Swin Transformer

Liu et al. proposed Swin Transformer, a hierarchical vision transformer based on shifted windows [5]. The model achieved 87.3% ImageNet-1K top-1 accuracy and strong dense-prediction results, including 53.5 mIoU on ADE20K validation in the original paper. Swin is relevant because it represents the transformer family and motivates SwinUNETR for segmentation and Swin-T as a classification comparison. In this project, Swin-T is treated as an important comparison experiment rather than the final selected classifier.

3.16 Improved ConvNeXt-Tiny and CT Classification Studies

Recent medical classification studies have also adapted ConvNeXt-Tiny for efficient medical-image recognition [11]. FocalNeXt further shows that ConvNeXt-related components can be integrated into CT classification systems [13]. These studies are not sinus-specific, but they support the use of lightweight modern convolutional backbones in limited medical datasets. This literature is consistent with the final project result, where ConvNeXt-Tiny gave better fold-level and out-of-fold performance than Swin-T.

3.17 2.5D Medical Imaging Approaches

Several medical-imaging studies use 2.5D representations to balance volumetric context and computational efficiency [15]. In this approach, representative slices or orthogonal views are used as multi-channel inputs for 2D pretrained networks. This strategy is relevant to the current project because each 3D ROI is converted into axial, coronal, and sagittal views. The resulting representation preserves partial 3D context while enabling transfer learning with pretrained

ConvNeXt-Tiny and Swin-T backbones.

3.18 State-of-the-Art Frameworks Analysis

The current state of the art related to this project can be understood as three connected directions. The first direction is anatomical sinus CT segmentation, where 3D CNNs, U-Net variants, SwinUNETR, and ensemble approaches have shown that sinonasal structures can be localized with useful accuracy. These studies are important because they support the feasibility of using segmentation masks as anatomical guidance.

The second direction is paranasal abnormality classification. Existing classification studies show that deep learning can detect abnormal maxillary sinus patterns, especially when self-supervised or contrastive learning is used to reduce dependence on large labeled datasets. However, these works are usually narrower than the present project because they focus mainly on maxillary anomaly classification rather than several co-existing sinus CT findings.

The third direction is the use of modern pretrained backbones and 2.5D medical-imaging representations. ConvNeXt, Swin Transformer, and 2.5D CT strategies are relevant because they provide practical feature-extraction methods for limited medical datasets. In this project, this direction motivated the transition from custom 3D ROI classifiers to pretrained 2.5D ConvNeXt-Tiny and Swin-T comparisons.

Based on these directions, the proposed system is positioned between segmentation-only and classification-only research. Its main goal is not to introduce a new segmentation architecture, but to connect anatomical segmentation with ROI-based multi-label abnormality prediction in sinus CT.

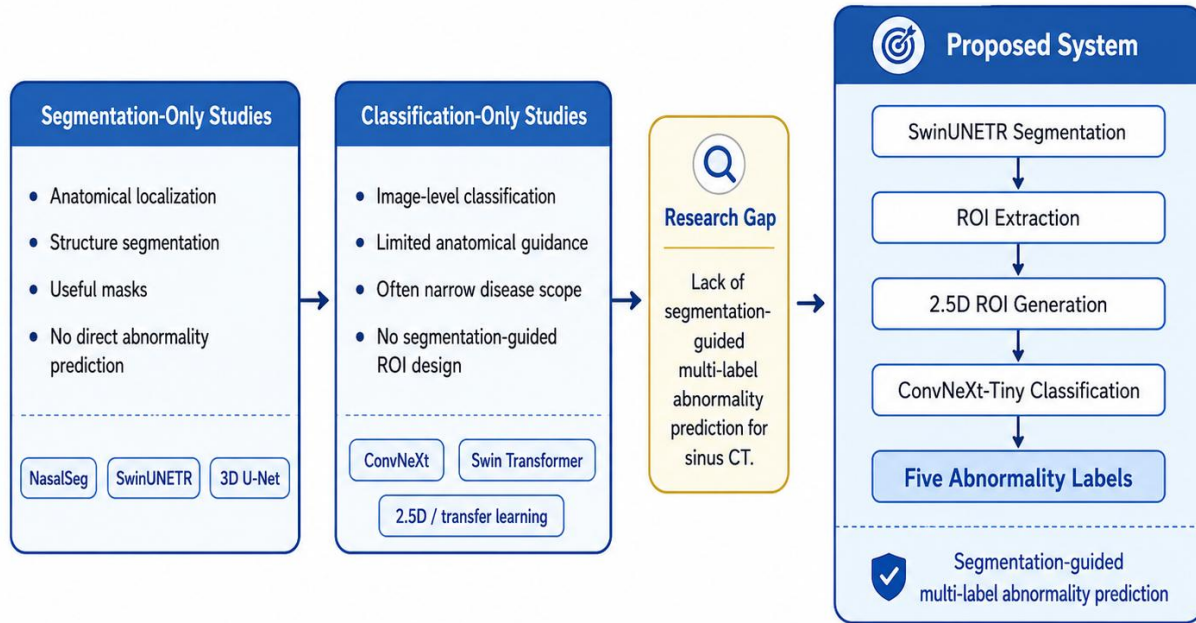


Figure 5: Research gap and positioning of the proposed system

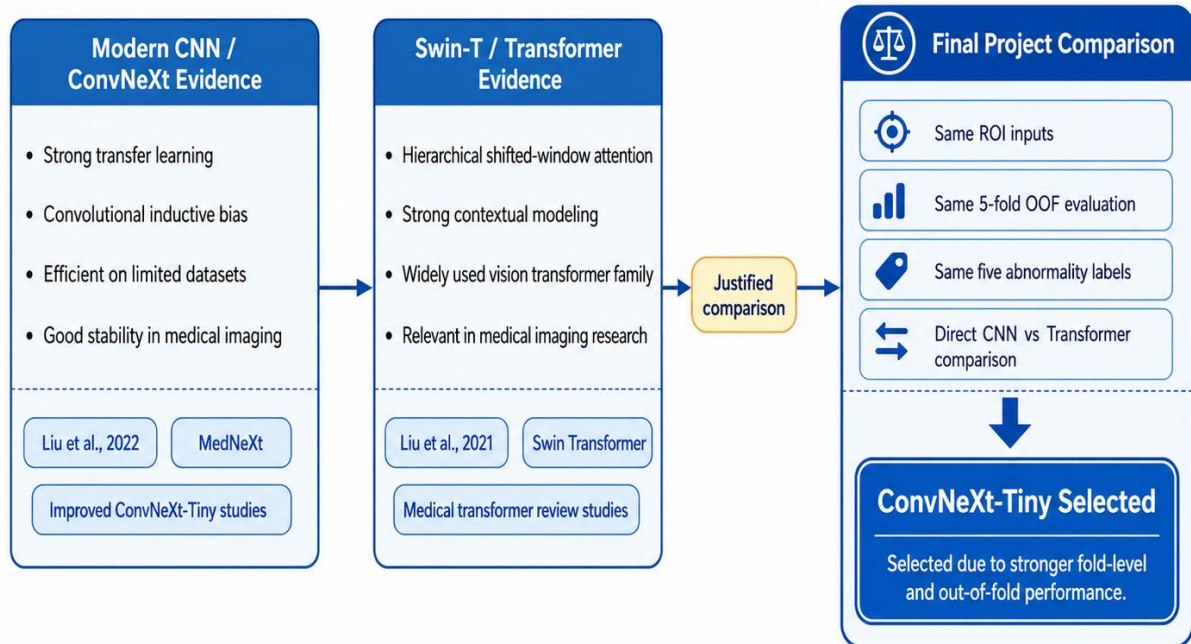


Figure 6: Literature-based justification of ConvNeXt-Tiny and Swin-T comparison

Table 1: Summary of related work in sinus CT analysis, backbones, and medical imaging

Paper	Approach	Architecture	Main	Strength	Limitation
Zhang et al., 2024, NasalSeg	Open-access CT dataset and benchmark for nasal cavity and paranasal sinus segmentation.	Dataset with voxel-level annotations; benchmark segmentation models.	Provides 130 3D CT scans with five annotated structures.	Public dataset directly related to this project.	Labels only five structures and do not directly include septum or turbinates.
Hatamizadeh et al., 2022, SwinUNETR	Transformer-based 3D medical image segmentation.	Swin Transformer encoder with U-shaped decoder.	Captures long-range context in 3D segmentation.	Strong architecture for volumetric medical segmentation.	Memory intensive and originally validated on other medical tasks.
Liu et al., 2022, ConvNeXt	Modernized convolutional architecture.	Pure ConvNet with transformer-era design improvements.	Shows that modern CNNs can compete with transformer backbones.	Supports choosing ConvNeXt-Tiny for limited-data transfer learning.	Designed for natural images; medical transfer still requires careful ROI design.
Liu et al., 2021, Swin Transformer	Hierarchical transformer with shifted windows.	Window-based self-attention and multi-scale features.	Efficient transformer backbone for vision tasks.	Good comparison model against ConvNeXt.	Can require more data and careful regularization.
Whangbo et al., 2024	Multi-class paranasal sinus segmentation in sinusitis CT.	3D U-Net variants.	Segments major sinus regions in CT images.	Clinically relevant paranasal sinus segmentation setting.	Focuses on segmentation rather than abnormality prediction.
Pani et al., 2025, 3D-NASE	Ensemble-based 3D nasal CT segmentation on NasalSeg.	3D U-Net, UNETR, and Swin UNETR with majority and soft voting.	Soft voting achieved Dice 94.50 ± 0.86 and mIoU 90.56 ± 0.78 .	Directly related to NasalSeg and combines CNN and transformer strengths.	Segmentation-only, computationally heavier, and does not address septum/turbinate abnormality classification.
	Self-	3D	Classifies	Addresses	Narrower

Bhattacharya et al., 2024	supervised learning for paranasal anomaly classification.	convolutional autoencoder / self-supervised pipeline.	normal versus anomalous maxillary sinus patterns.	limited labeled data through self-supervision.	classification scope than multi-label abnormality prediction.
Ozturk et al., 2024	Automatic maxillary sinus segmentation on CBCT.	U-Net segmentation model.	Segments maxillary sinus structures.	Relevant to maxillary anatomy and dental/sinus imaging.	CBCT-focused and not a full multi-label CT abnormality pipeline.
Song et al., 2025	Comparison of CNNs, ViTs, and hybrid networks for sinus segmentation.	CNN, ViT, and hybrid segmentation networks.	Compares architecture families for paranasal sinus segmentation.	Useful for understanding architecture trade-offs.	Segmentation-focused and does not solve clinical multi-label classification.
Xia et al., 2025	Medical image classification using an improved ConvNeXt-Tiny.	Lightweight ConvNeXt-Tiny with pooling and attention refinements.	Shows ConvNeXt-Tiny can be adapted to medical classification.	Supports ConvNeXt-Tiny as a reasonable classification backbone.	Not specific to sinus CT or multi-label ROI classification.
Roy et al., 2023, MedNeXt	Medical image architecture based on ConvNeXt-style blocks.	3D ConvNeXt encoder-decoder.	Adapts ConvNeXt principles to medical imaging.	Supports modern CNNs in data-scarce medical settings.	Segmentation-focused, not a direct classifier for sinus CT.
Gulsoy and Kablan, 2025, FocalNeXt	CT lung cancer classification using ConvNeXt-augmented architecture.	ConvNeXt + FocalNet hybrid.	Applies ConvNeXt-related features to CT classification.	Shows ConvNeXt components are useful in CT classification.	Different disease and dataset.
Aburass et al., 2025	Review of vision transformers in medical imaging.	Review of ViT/Swin-style methods.	Summarizes transformer applications in medical classification and segmentation.	Justifies including Swin-T as a comparison.	Review-level evidence; not a sinus-specific experiment.

Wang et al., 2024	2.5D deep learning and conventional features for CT-based prediction.	2.5D masks/features plus model fusion.	Uses 2.5D representation to balance context and efficiency.	Supports 2.5D design for limited data.	Different clinical target and not sinus CT.
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3.19 Comparative Analysis of Existing Systems

Existing work related to this project can be divided into three main groups: sinus CT segmentation studies, maxillary sinus abnormality classification studies, and general medical-image backbone studies. Direct comparison is difficult because these studies do not solve exactly the same task. Some focus only on anatomical segmentation, some classify only maxillary sinus anomalies, and others introduce general architectures rather than sinus-specific systems. However, they are still useful for understanding how the proposed system differs from previous approaches.

Table 2: Comparison of existing approaches and the proposed system

Study / Approach	Main task	Model / Method	Reported result / focus	Strength	Limitation compared with this project
Zhang et al., NasalSeg [1]	Nasal cavity and paranasal sinus segmentation	Dataset and benchmark segmentation models	130 CT scans with five annotated sinonasal structures	Directly related dataset and anatomical labels	Does not provide abnormality classification or direct septum/turbinate masks
Whangbo et al. [6]	Multi-class paranasal sinus segmentation	3D U-Net variants	Best reported F1 of 84.29% on normal test set and 79.32% on abnormal test set	Clinically relevant sinusitis CT segmentation	Segmentation-only; no multi-label abnormality prediction
Song et al. [10]	Paranasal sinus segmentation comparison	CNN, ViT, and hybrid models	Swin UNETR achieved Dice 0.830 and Jaccard 0.719	Shows value of hybrid/transformer segmentation in sinus CT	Does not address classification of abnormalities
Ozturk et al. [9]	Maxillary sinus segmentation	U-Net on CBCT	Automatic maxillary sinus localization	Relevant to maxillary sinus anatomy	CBCT-focused and not a full sinus CT multi-label system

Bhattacharya et al. [7]	Maxillary anomaly classification	Self-supervised 3D convolutional model	AUPRC 0.79 using 10% annotated data	Handles limited annotation for anomaly classification	Focuses only on maxillary anomaly, not five-label sinus CT classification
Bhattacharya et al. [8]	Maxillary anomaly classification	Supervised contrastive 3D CNN	AUROC 0.85 compared with 0.66 for cross-entropy baseline	Improves representation learning for limited data	Narrower task than the proposed multi-label setting
Proposed system	Segmentation-guided sinus CT abnormality classification	SwinUNETR + ROI extraction + ConvNeXt-Tiny	Segmentation Dice 0.8743; ConvNeXt-Tiny OOF macro F1 0.6928	Combines anatomical localization with five-label abnormality prediction	Limited by 130 cases, class imbalance, and missing direct septum/turbinate masks

Table 3: Summary of Reported Results in Related Studies

Paper	Iou / mIoU	Dice	F1	Accuracy	Precision	Recall
Pani et al. 3D-NASE 2025	90.56	94.50	-	99.70	94.27	-
Swin UNETR, Hatamizadeh et al., 2022	-	89.1	-	-	-	-
Swin Transformer, Liu et al., 2021	53.5	-	-	-	-	-
Whangbo et al., 2024	-	-	F1 84.29% normal, 79.32% abnormal	-	-	-
Ozturk et al., U-Net maxillary sinus segmentation, 2024	92.75	-	97.84	-	-	-
Song et al., CNN/ViT/hybrid sinus segmentation, 2025	71.9	83.0	-	-	93.5	75.8
Improved ConvNeXt-Tiny, Xia et al., 2025	-	-	-	89.10	-	-
MedNeXt, Roy et al., 2023	-	89.22	-	-	-	-
FocalNeXt, Gulsoy and Kablan, 2025	-	-	99.56	99.81	-	99.36

3.20 Research Gap and Proposed Innovation

The reviewed studies show that sinus CT analysis has been studied mainly from two separate

directions. The first direction focuses on anatomical segmentation, where models such as U-Net variants and SwinUNETR can localize sinus and nasal cavity structures. These studies are important, but they usually stop at segmentation and do not directly predict clinical abnormality labels.

The second direction focuses on abnormality classification, especially maxillary sinus anomaly detection. These studies show that deep learning can recognize abnormal findings in paranasal imaging, but most of them solve a narrower problem than the one addressed in this project. They usually classify maxillary sinus anomaly only, rather than predicting several co-existing abnormalities from the same CT case.

Therefore, the main research gap is the lack of a complete segmentation-assisted multi-label pipeline for sinus CT abnormality prediction. In the proposed problem, the model must predict five labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. This is more difficult than single-label classification because these findings may co-occur in one patient.

The proposed innovation is to connect anatomical segmentation with abnormality classification. SwinUNETR is used to localize available sinonasal structures, then these structures guide ROI extraction for ConvNeXt-Tiny classification. This makes the system more interpretable than a full-volume classifier because each prediction is linked to an anatomical region. The project also introduces contextual ROI handling for labels without direct masks: the septum is represented using both nasal cavities, and inferior turbinate hypertrophy is represented using the ipsilateral nasal cavity.

The proposed innovation should therefore be understood as a workflow-level contribution. The project does not only test a classifier on sinus CT images; it designs a complete pipeline where available anatomical masks determine the classification inputs. This is important because the clinical findings are region-dependent and because the available segmentation labels do not perfectly match all diagnostic targets.

3.21 Chapter Summary

This chapter reviewed the literature related to sinonasal CT segmentation, paranasal abnormality classification, modern image-classification backbones, and 2.5D medical-imaging strategies. The

reviewed studies show that anatomical segmentation in sinus CT is feasible and that abnormality classification has been explored mainly in narrower maxillary-focused tasks. However, fewer studies connect segmentation-guided ROI extraction with multi-label sinus CT abnormality prediction. This gap motivates the proposed pipeline, where SwinUNETR provides anatomical localization and ConvNeXt-Tiny performs ROI-based classification for five selected abnormality labels.

Chapter Four: Research Methodology

4.1 Introduction

This chapter describes the complete methodology used to build the proposed AI system for sinus CT analysis. The system was designed as a segmentation-assisted multi-label classification pipeline rather than a direct full-volume classifier. This design was selected because the target abnormalities are anatomically localized: maxillary sinus abnormalities are related to the maxillary sinuses, nasal septum abnormality is related to the central region between the nasal cavities, and inferior turbinate hypertrophy is related to the nasal cavity regions.

The methodology consists of five main stages. First, CT volumes and anatomical masks are preprocessed to standardize orientation, spacing, intensity range, and tensor format. Second, a 3D SwinUNETR model is trained to segment the available NasalSeg structures. Third, the predicted or reference segmentation masks are used to extract anatomy-guided regions of interest. Fourth, each 3D ROI is converted into a 2.5D representation using axial, coronal, and sagittal views. Finally, a ConvNeXt-Tiny classifier predicts five abnormality labels using independent sigmoid outputs.

This design makes the classification stage more clinically meaningful than a blind image-level classifier. Instead of forcing the model to search the entire CT volume for every label, each prediction is connected to a specific anatomical region. This is especially important in sinus CT analysis because abnormalities may co-occur in the same patient, and some target structures, such as the nasal septum and inferior turbinates, are not directly segmented in the available dataset.

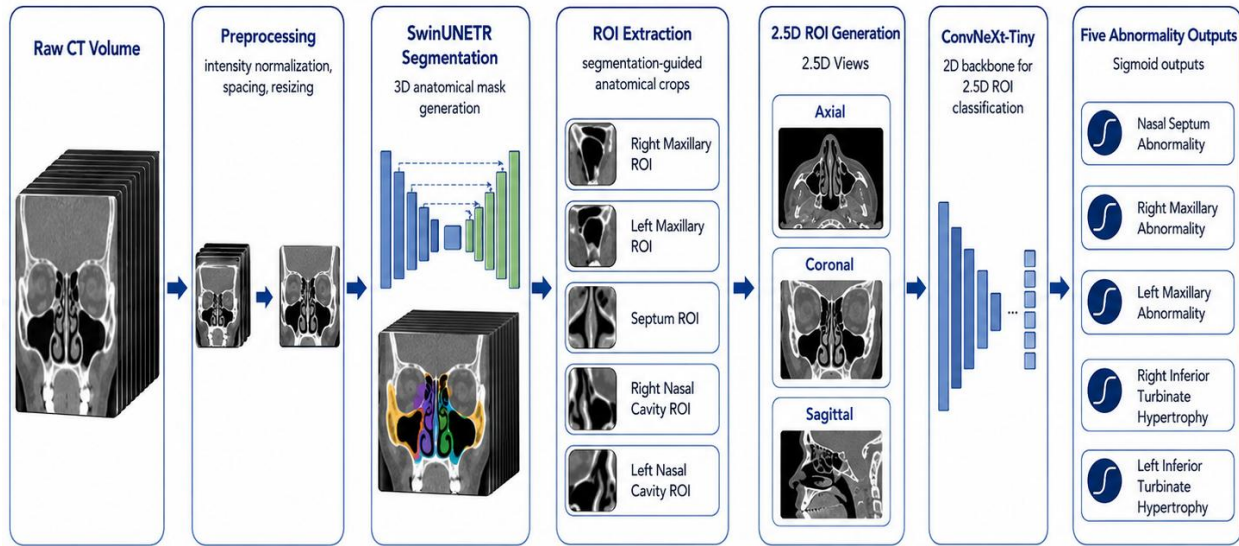


Figure 7: Overall pipeline of the proposed sinus CT analysis system

4.1.1 Classical and Baseline Approaches for Sinus CT Analysis

Before selecting the final deep-learning pipeline, several simpler approaches can be considered for sinus CT analysis. These approaches are useful as conceptual baselines, but they are limited for the specific objective of this project: segmentation-assisted multi-label abnormality prediction from sinus CT scans.

A traditional clinical approach is manual CT interpretation by a radiologist or ENT specialist. This remains the clinical reference standard because sinus CT findings require anatomical knowledge and clinical judgment. However, manual interpretation is time-consuming, depends on the reader’s experience, and may vary between observers. This project does not aim to replace expert interpretation, but to build a research prototype that can assist anatomical localization and abnormality prediction.

Classical image-processing methods can also be considered for sinus CT because air-filled cavities and bony structures have different Hounsfield Unit ranges. Thresholding, region growing, connected-component analysis, and morphological operations may help isolate air cavities such as the maxillary sinuses. However, these methods are sensitive to anatomical variation, mucosal thickening, partial opacification, and irregular sinus boundaries. They are also not sufficient for predicting clinical abnormality labels such as septum abnormality or inferior turbinate hypertrophy.

Classical machine-learning methods, such as support vector machines or random forests, could be trained using handcrafted features extracted from sinus regions. Possible features include intensity statistics, texture descriptors, volume measurements, or shape features. However, this requires manually designing features for each anatomical target. In this project, the abnormalities are region-dependent, multi-label, and sometimes indirectly represented because the dataset does not provide direct septum or turbinate masks. Therefore, handcrafted features alone are unlikely to capture the full appearance variation needed for reliable prediction.

A direct full-volume deep-learning classifier is another possible baseline. In this approach, the complete CT volume or selected slices are passed to a classifier without anatomical localization. Although this is simpler, it is less suitable for this project because the target labels are connected to specific anatomical regions. A full-volume classifier may learn irrelevant correlations or focus on dominant structures while missing rare labels such as maxillary abnormalities.

For these reasons, the proposed method uses a segmentation-assisted pipeline. SwinUNETR first localizes the available sinonasal structures, then anatomy-guided ROIs are extracted and passed to a multi-label classifier. This design is more appropriate for the medical nature of the problem because it connects each prediction to a clinically meaningful region while reducing irrelevant CT information before classification.

4.2 Dataset Source and Annotation Story

The dataset preparation stage was a critical part of this project because the available CT volumes, anatomical masks, and diagnostic annotations were not directly ready for classification. The final classification task was not obtained by simply training on all available annotation columns. Instead, the raw information had to be reviewed, filtered, and reformulated according to the anatomical masks available in the NasalSeg dataset and the amount of positive data available for each finding.

The final dataset used in this project was therefore produced through three connected steps. First, CT volumes and anatomical segmentation masks were obtained from the NasalSeg dataset. Second, the medical annotation workbook was reviewed and converted into target labels. Third, only the findings that were medically meaningful, sufficiently represented, and anatomically connected to the available masks were retained for the final classification task.

4.2.1 Data Source and Acquisition

The imaging data used in this project were based on the NasalSeg CT dataset. The dataset contains 130 sinus CT cases with voxel-wise anatomical masks. These CT volumes provided the imaging input for the segmentation stage and the anatomical basis for ROI extraction in the classification stage.

The classification stage also used a clinical annotation workbook prepared with medical input. This workbook contained case-level descriptions of sinonasal findings across several anatomical regions. Unlike the segmentation masks, these annotations were not voxel-wise labels. They described the presence and type of findings for each case and therefore required cleaning before they could be used as supervised classification targets.

The CT cases, segmentation masks, and annotation workbook were matched at the case level. Only matched cases were used in the final experiments so that each CT volume had both anatomical localization information and corresponding diagnostic labels.

4.2.2 Available Segmentation Masks

The NasalSeg segmentation labels include five anatomical structures: the right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. These masks were used for anatomical localization and ROI extraction. They were not diagnostic labels by themselves.

The available masks strongly affected the classification design. Right and left maxillary sinus abnormalities could be connected directly to the corresponding maxillary sinus masks. However, the dataset did not provide direct masks for the nasal septum, inferior turbinates, or middle turbinates. Therefore, any diagnostic target related to these structures had to be handled using contextual anatomical regions or excluded if the available data were too limited.

Table 4: Dataset and annotation sources used in the project

Component	Source / form	Role in the project
CT volumes	NasalSeg dataset	Input imaging data
Segmentation masks	NasalSeg voxel-wise masks	Anatomical localization and

		ROI extraction
Available mask structures	Right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, nasopharynx	Defines which anatomical ROIs can be extracted
Medical annotation workbook	Case-level clinical annotations	Source of abnormality labels
Final matched cases	130 CT cases	Used for segmentation and classification experiments

4.2.3 Medical Annotation Workbook

The medical annotation workbook contained more detailed information than the final classification model could directly use. For example, the nasal septum annotations included several descriptions such as mild, moderate, or severe deviation and direction of deviation. The maxillary sinus annotations also included different descriptions such as partial opacification, retention cyst, hypoplasia, or complete opacification.

These detailed descriptions were medically useful, but they created a learning problem when used as separate classes. The dataset contained only 130 cases, and many detailed values appeared only a few times. Training a multi-class subtype classifier under these conditions would create very small subclasses and unreliable evaluation. Therefore, the annotations were not used as detailed subtype classes. Instead, they were converted into binary presence-or-absence labels for selected anatomical targets.

4.2.4 Annotation Cleaning and Usable-Target Filtering

The raw annotation workbook contained more information than could be safely used in the final model. Some findings were not connected to anatomical regions available in the segmentation masks, while others had too few positive cases to support reliable supervised learning. Since the proposed pipeline depends on segmentation-guided ROIs, the final labels had to satisfy two conditions.

First, each selected label had to be medically connected to an available or justifiable ROI. For example, maxillary sinus labels were directly connected to maxillary sinus masks. Nasal septum abnormality was connected to a contextual ROI derived from both nasal cavities because the

septum lies between them. Inferior turbinate hypertrophy was connected to the ipsilateral nasal cavity ROI because the inferior turbinate lies inside the nasal cavity.

Second, each selected label needed enough positive examples to make training and evaluation meaningful. Labels with extremely small positive counts were not retained because they would make the classifier unstable and could produce misleading metrics. This filtering step was necessary because the goal was to build a reliable five-label research prototype rather than to force the model to learn every rare annotation value in the workbook.

Table 5: Annotation groups and filtering decisions

Annotation group	Raw annotation detail	Positive count	Final decision	Reason
Nasal septum	Multiple deviation descriptions	71	Used	Clinically important and connected to bilateral nasal cavity context
Right maxillary sinus	Multiple abnormal descriptions	22	Used	Direct right maxillary sinus mask available
Left maxillary sinus	Multiple abnormal descriptions	23	Used	Direct left maxillary sinus mask available
Right inferior turbinate	Hypertrophy	60	Used	Connected to right nasal cavity ROI
Left inferior turbinate	Hypertrophy	60	Used	Connected to left nasal cavity ROI
Right middle turbinate	Concha bullosa	13	Excluded	Too few positive cases and no direct mask
Left middle turbinate	Concha bullosa / hypertrophy	12	Excluded	Too few positive cases and no direct mask

4.2.5 Rare-Label Reduction and Middle Turbinate Exclusion

One of the main decisions in dataset preparation was removing the middle turbinate labels from the final classification task. The annotation workbook showed only 13 positive cases for the right middle turbinate and 12 positive cases for the left middle turbinate. These counts were much smaller than the already limited dataset size and would make model training and evaluation unstable.

The middle turbinate labels were also not supported by direct segmentation masks. Since the dataset did not include a middle turbinate mask, the model would have needed to infer this

finding from a wider contextual nasal cavity region. Combining very low positive counts with indirect anatomical localization would increase the risk of unreliable predictions and near-zero F1 scores.

For this reason, the final classification task was reduced from the broader raw annotation set to five more stable labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy. This reduction was not a loss of medical relevance; it was a necessary design decision to keep the supervised learning task measurable and scientifically defensible.

4.2.6 Final Binary Multi-Label Formulation

After annotation cleaning and rare-label reduction, the final classification task was formulated as a five-label binary multi-label problem. Each selected target was represented as present or absent. This means that the model did not attempt to predict detailed subtypes such as the direction or severity of septal deviation, or the exact subtype of maxillary sinus abnormality. Instead, it predicted whether an abnormality was present in each selected anatomical target.

This binary formulation was selected because the dataset size was limited and many detailed annotation values appeared only a few times. For example, several septum deviation descriptions occurred in fewer than five cases, and some maxillary descriptions occurred only once. Using these values as separate classes would create unreliable subclasses and would make the evaluation less meaningful.

The final task remained multi-label at the CT-case level because more than one abnormality could be present in the same patient. Therefore, each case was represented by a vector of five binary targets. A case without any of the selected final abnormality labels was represented as an all-zero vector for this task. This all-zero vector means that none of the five selected targets is positive; it should not be interpreted as a complete radiological normality statement if other excluded findings exist outside the final label set.

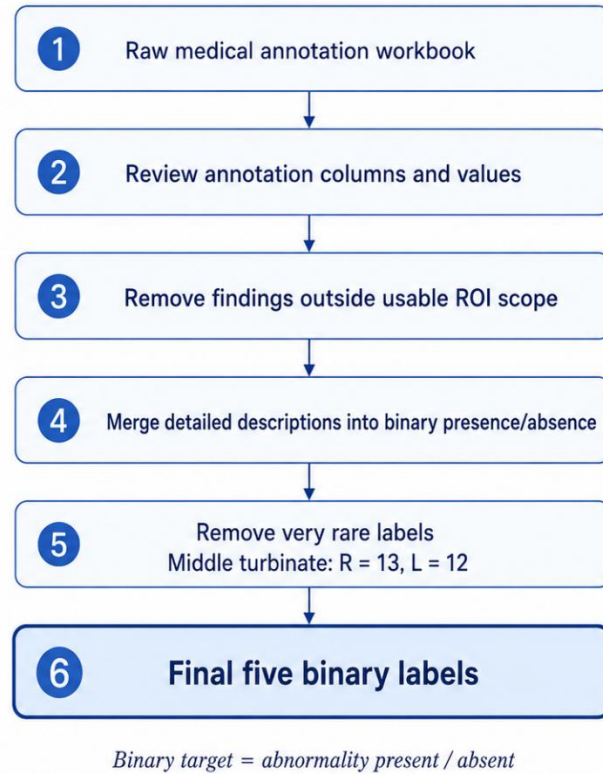


Figure 8: Label preparation workflow from raw annotations to final binary targets

4.3 Data Preprocessing

Data preprocessing was performed separately for the segmentation and classification stages because each stage uses a different input representation. The segmentation stage works directly on 3D CT volumes, while the classification stage uses cropped anatomy-guided ROIs converted into 2.5D images.

For the segmentation stage, CT volumes were loaded as 3D medical images and processed using MONAI transforms. The main preprocessing operations included orientation handling, spacing normalization, intensity windowing, intensity scaling, and conversion to tensors. Spacing normalization was used to reduce variation in voxel size between CT cases. The selected input spacing was $1.0 \times 1.0 \times 1.5$ mm, which provides a balance between anatomical detail and computational memory cost. The HU window was set to -400 to 1000. This range preserves useful sinonasal air, soft-tissue, and bony information while reducing the effect of extreme intensity values outside the relevant anatomical range.

For the classification stage, preprocessing started after the CT volume and its corresponding

segmentation mask were matched. The segmentation mask was used to define the anatomical ROI for each target label. Each extracted ROI contained the CT intensity crop and anatomical mask context. The purpose of this design was to provide the classifier with both appearance information and localization information. The CT crop represents the visual pattern of the tissue, while the mask context informs the model about the anatomical region being analyzed.

After ROI extraction, each crop was resized to a fixed spatial size to ensure that all samples had a consistent input format. This is necessary because sinus anatomy differs between patients and because the bounding boxes extracted from masks may have different sizes. The resized ROI was then converted into a 2.5D image using representative axial, coronal, and sagittal views. These three views were stacked as channels to create an input compatible with pretrained two-dimensional classification backbones.

This preprocessing strategy was chosen to reduce unnecessary anatomical variation and to focus the classifier on clinically relevant regions. It also reduced memory requirements compared with training a full 3D classifier on entire CT volumes, which is important because the dataset contains only 130 matched CT cases.

4.4 Data Augmentation

Data augmentation was applied only to the training data in order to improve robustness without contaminating validation or test evaluation. Validation and test samples were processed using deterministic transforms only. This ensured that reported metrics reflected model generalization rather than random augmentation effects.

For segmentation, augmentation was applied to the 3D CT volumes and masks using spatial and intensity transformations. Spatial augmentation included random cropping, flipping, and rotation. These operations help the model become more robust to small differences in patient positioning and anatomical shape. Intensity augmentation included Gaussian noise and intensity shift or scale transformations, which help the model handle variation in CT appearance and acquisition conditions.

For classification, augmentation was applied to the 2.5D ROI images rather than to the full CT volume. The goal was to improve robustness of the ROI classifier while preserving the clinical label. Light augmentation was preferred because strong geometric transformations can distort

anatomical orientation, which is important in sinus CT. The final augmentation strategy avoided creating saved duplicate ROI images because earlier experiments showed that saved augmented copies mainly produced near-duplicate versions of the same patients. Instead, light on-the-fly augmentation was used during training.

This augmentation strategy reflects the medical nature of the problem. In natural image classification, aggressive augmentation can often be useful, but in medical CT analysis, excessive rotation, cropping, or distortion can remove clinically meaningful spatial relationships. Therefore, augmentation was used conservatively to improve generalization while preserving anatomical validity.

4.5 Data Partitioning and 5-Fold Evaluation

The segmentation notebook used an executed split of 91 training cases, 19 validation cases, and 20 test cases. The classification notebook used balanced 5-fold out-of-fold evaluation across the 130 matched cases. This setup ensured that each case was evaluated by a model that did not train on it. The 5-fold design was selected because the dataset is limited and a single train/test split could produce unstable results, especially for rare labels such as maxillary abnormalities.

Table 6: Data partitioning strategy used in this project

Stage	Partitioning	Purpose
Segmentation	91 train / 19 validation / 20 test	Train SwinUNETR and evaluate anatomical segmentation on held-out cases.
Classification	Balanced 5-fold OOF	Compare classifiers while using every case once as validation.
Threshold tuning	OOF predictions	Select label-specific probability thresholds for final metrics.

4.6 SwinUNETR Segmentation Stage

The segmentation stage uses SwinUNETR, a 3D transformer-based U-shaped architecture. The model is trained to predict six classes: background, right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx. The training objective is DiceCE loss, which combines overlap-based Dice loss and cross-entropy loss. This is suitable for segmentation because it encourages both voxel-level class prediction and mask overlap.

Table 7: SwinUNETR segmentation training configuration

Parameter	Value
Architecture	SwinUNETR
Framework	MONAI / PyTorch
Input spacing	$1.0 \times 1.0 \times 1.5$
ROI size	$96 \times 96 \times 32$
HU window	-400 to 1000
Loss function	DiceCE loss
Optimizer	AdamW
Validation method	Sliding-window inference
Executed split	91 train / 19 validation / 20 test cases

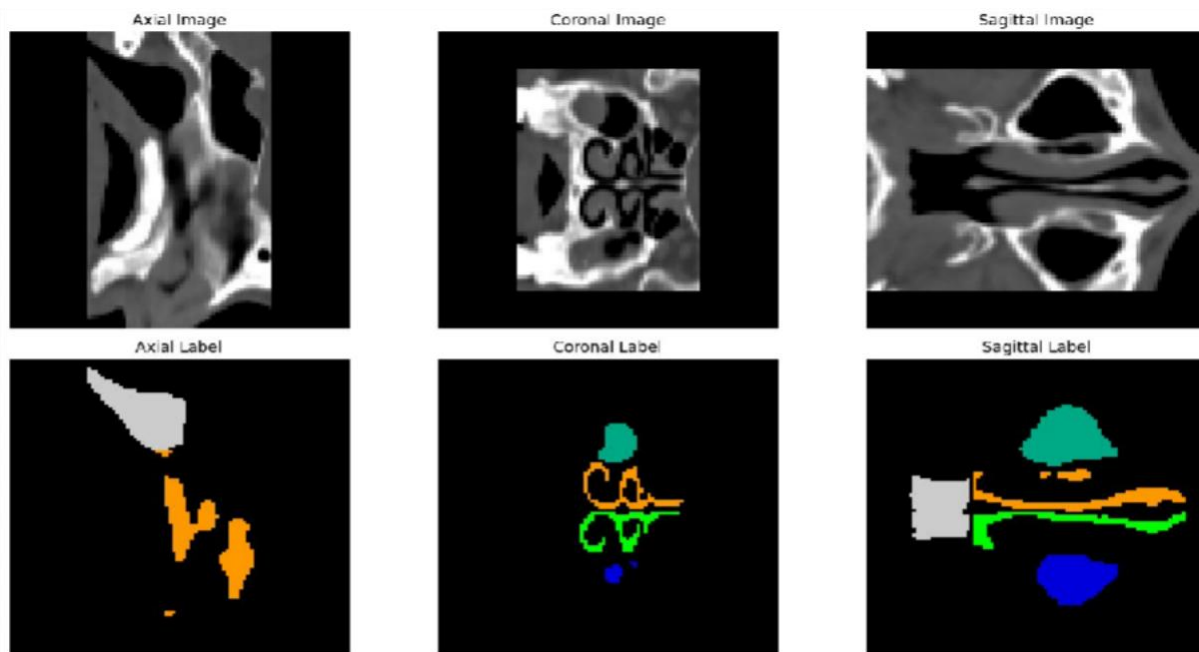


Figure 9: Augmentation sanity-check visualization

4.7 ROI Extraction Strategy

The final classification model uses five stable contextual ROIs. The right and left maxillary labels use the corresponding maxillary sinus masks directly. The nasal septum label uses a contextual ROI derived from both nasal cavities with padding. This design follows the anatomical fact that the nasal septum lies between the two nasal cavities, while avoiding brittle pseudo-septum masks. The inferior turbinate labels use the ipsilateral nasal cavity ROIs because direct turbinate masks are not available in the dataset.

Table 8: ROI-to-label mapping used in the final classification system

Target label	ROI name	Segmentation labels used	Reasoning
right_maxillary_abnormal	right_maxillary_roi	Right maxillary sinus	Direct anatomical match.
left_maxillary_abnormal	left_maxillary_roi	Left maxillary sinus	Direct anatomical match.
nasal_septum_abnormal	nasal_septum_region_roi	Right + left nasal cavities	Septum lies between both nasal cavities; contextual ROI is more stable.
right_inferior_turbinate_abnormal	right_nasal_cavity_roi	Right nasal cavity	Inferior turbinate is located inside the nasal cavity; no direct turbinate mask available.
left_inferior_turbinate_abnormal	left_nasal_cavity_roi	Left nasal cavity	Inferior turbinate is located inside the nasal cavity; no direct turbinate mask available.

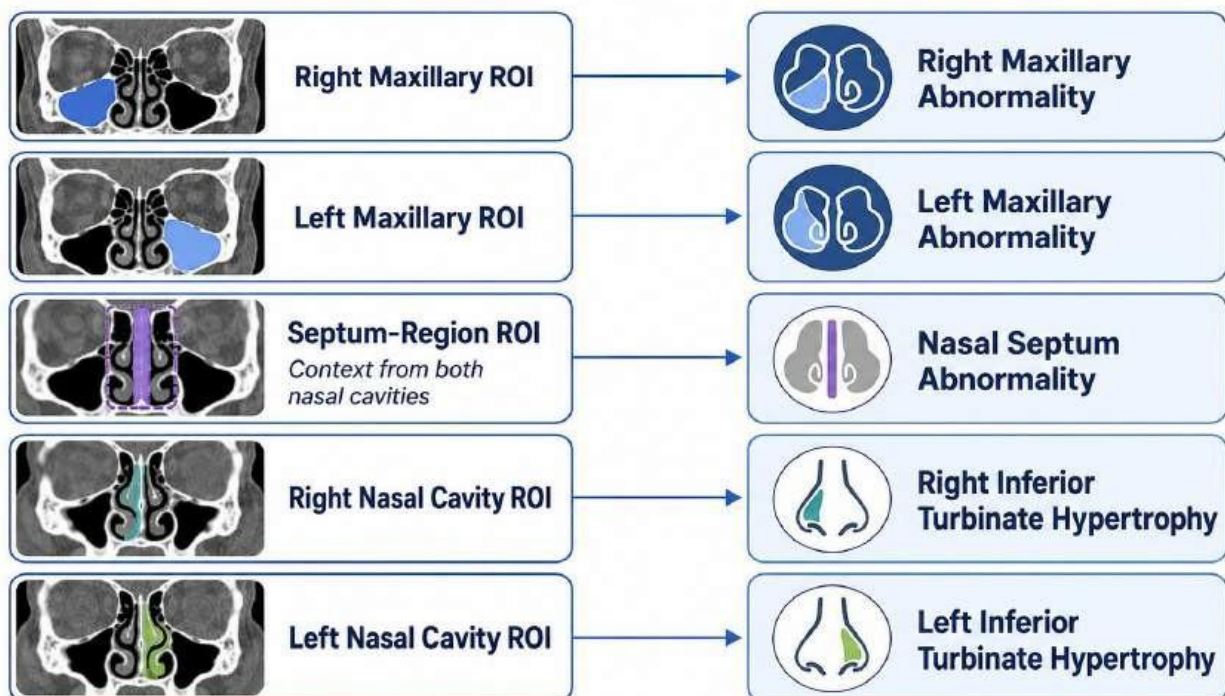


Figure 10: Final ROI definitions used in the classification stage

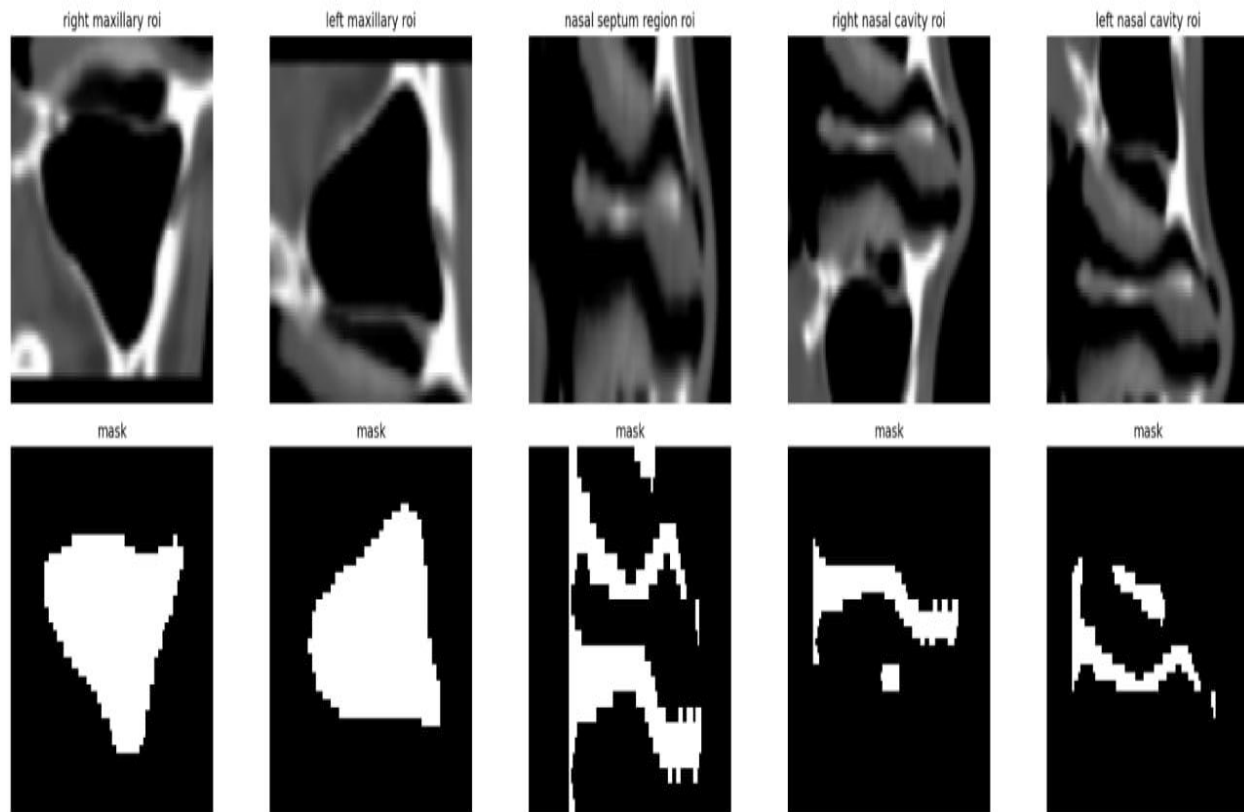


Figure 11: Example ROI extraction results from the classification notebook

4.8 2.5D ROI Generation

Each 3D ROI was converted into a 2.5D representation by taking representative axial, coronal, and sagittal views. These three views were stacked as a three-channel image. This design preserved partial volumetric context while allowing the use of pretrained 2D image classification backbones. It also reduced the memory cost compared with fully 3D classifiers, which is important for a dataset of 130 cases.

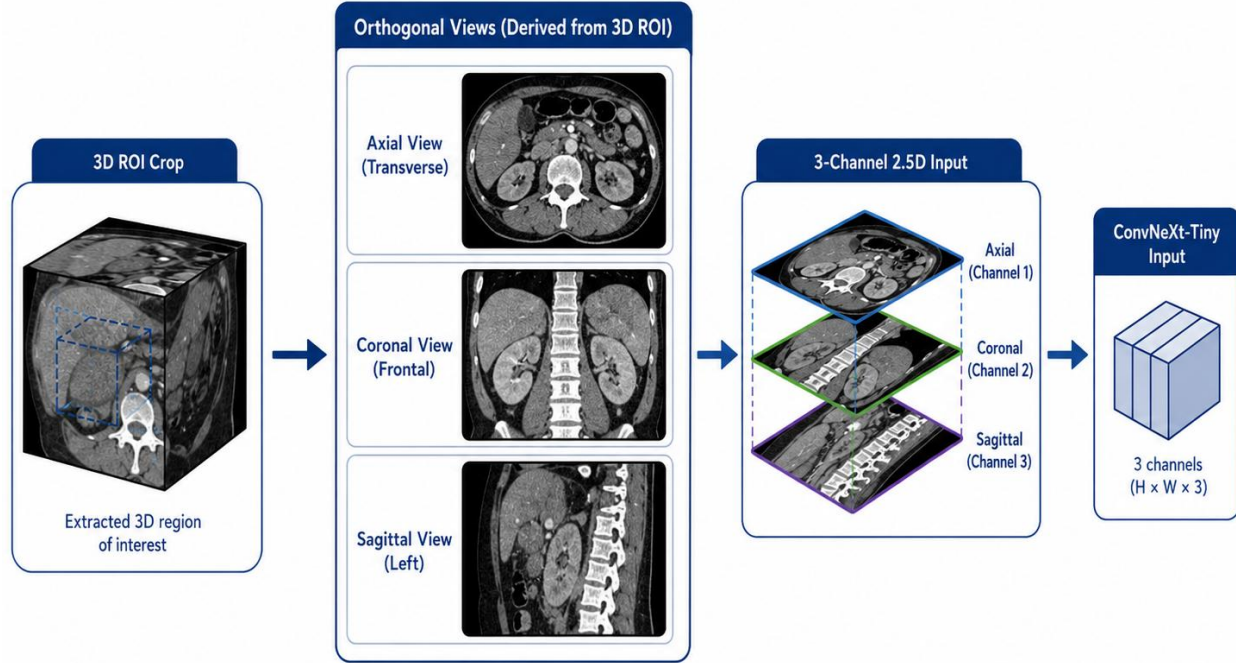


Figure 12: 2.5D ROI generation from axial, coronal, and sagittal views

4.9 Classification Input-Output Formulation and Case-Level Assembly

The classification stage is designed at the CT-case level, but its inputs are generated from anatomical ROIs. For each CT case, the segmentation masks are used to extract label-related ROIs. Each ROI is then converted into a 2.5D image by stacking representative axial, coronal, and sagittal views.

The final five labels are not treated as five unrelated image classes. Instead, each label is connected to a medically relevant ROI. The right maxillary abnormality label is linked to the right maxillary sinus ROI, the left maxillary abnormality label is linked to the left maxillary sinus ROI, the nasal septum abnormality label is linked to a bilateral nasal cavity context ROI, and the inferior turbinate labels are linked to the corresponding right or left nasal cavity ROIs.

$$\hat{y}_{case} = [\hat{y}_{septum}, \hat{y}_{R-maxillary}, \hat{y}_{L-maxillary}, \hat{y}_{R-turbinate}, \hat{y}_{L-turbinate}]$$

Equation 11: Case-level multi-label prediction vector

Therefore, the classifier should be understood as an ROI-based binary decision system whose outputs are assembled into a final case-level multi-label vector. Conceptually, the system evaluates the relevant ROI information for each label and then combines the five binary

decisions into one output vector for the CT case.

During implementation, the ROI samples can be processed one by one or batched together for computational efficiency. If the ROIs are batched, the batch dimension contains separate ROI images; the five ROIs are not merged into a single multi-channel image. Each 2.5D ROI image still contains three channels: axial, coronal, and sagittal.

This formulation avoids the weakness of a blind full-volume classifier. The model does not need to search the entire CT scan for every abnormality. Instead, each abnormality prediction is guided by the anatomical region most relevant to that label.

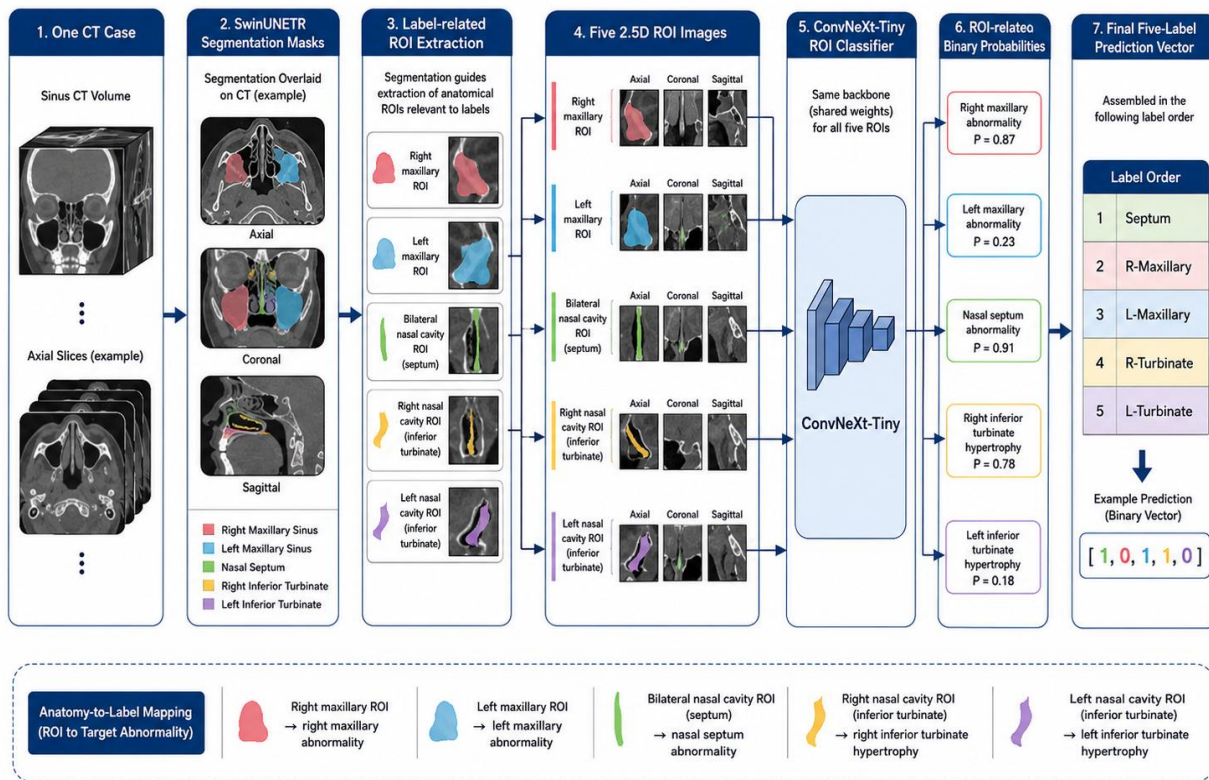


Figure 13: Case-level ROI Classification and Final Label Assembly

4.10 Important Exploratory Experiments

The classification stage went through several major experimental phases before reaching the current ConvNeXt-Tiny result. This section documents the important development path because the final system was not obtained from a single direct experiment. The project first established anatomical segmentation, then tested ROI-based 3D classification, then moved toward 2.5D transfer learning after several limitations became clear.

The first foundation was the SwinUNETR segmentation model. It segmented the available NasalSeg anatomical structures and provided the masks used to crop regions of interest. This changed the classification task from blind whole-volume prediction into anatomy-guided prediction, where each label could be linked to a clinically meaningful region.

After segmentation, the diagnosis problem was reformulated as multi-label classification. This was necessary because a single sinus CT case can contain septal abnormality, turbinate hypertrophy, and maxillary sinus abnormality at the same time. The model therefore used independent sigmoid outputs rather than a single softmax class.

The first diagnostic models used segmentation-guided 3D ROI crops with a ResNet-style classifier. Early experiments included seven abnormality labels, but the middle turbinate labels had too few positive cases and produced unstable zero or near-zero F1 scores. Therefore, the target set was reduced to the five main labels used in the final system.

The 3D ROI classifier was then improved using two-channel inputs. Each ROI contained both the CT intensity crop and the corresponding mask crop, allowing the model to learn image appearance and anatomical location together. Separate label-specific heads were also more stable than shared bilateral heads, because shared turbinate heads tended to increase false positives.

Several imbalance and augmentation strategies were tested. Raw positive class weighting was too aggressive, so softened square-root positive weights and capped weights were explored. Light on-the-fly CT intensity augmentation was useful, while saved augmented ROI copies were rejected because they mostly created near-duplicate versions of the same patients rather than true new clinical diversity.

Threshold tuning also became necessary. A fixed threshold of 0.5 was not reliable for all labels because each abnormality had a different prevalence and probability distribution. Hybrid and validation-based thresholds improved some fixed-split results, but the results remained sensitive to the split, which motivated the move toward 5-fold out-of-fold evaluation.

The best fixed-split ROI-based 3D ResNet model achieved an approximate macro F1 around 0.63, but later cross-validation showed that this was likely optimistic. In the more honest 5-fold evaluation, the ROI 3D ResNet approach reached an out-of-fold macro F1 of approximately

0.588, confirming that the custom 3D classifier had moderate capability but also a clear performance ceiling.

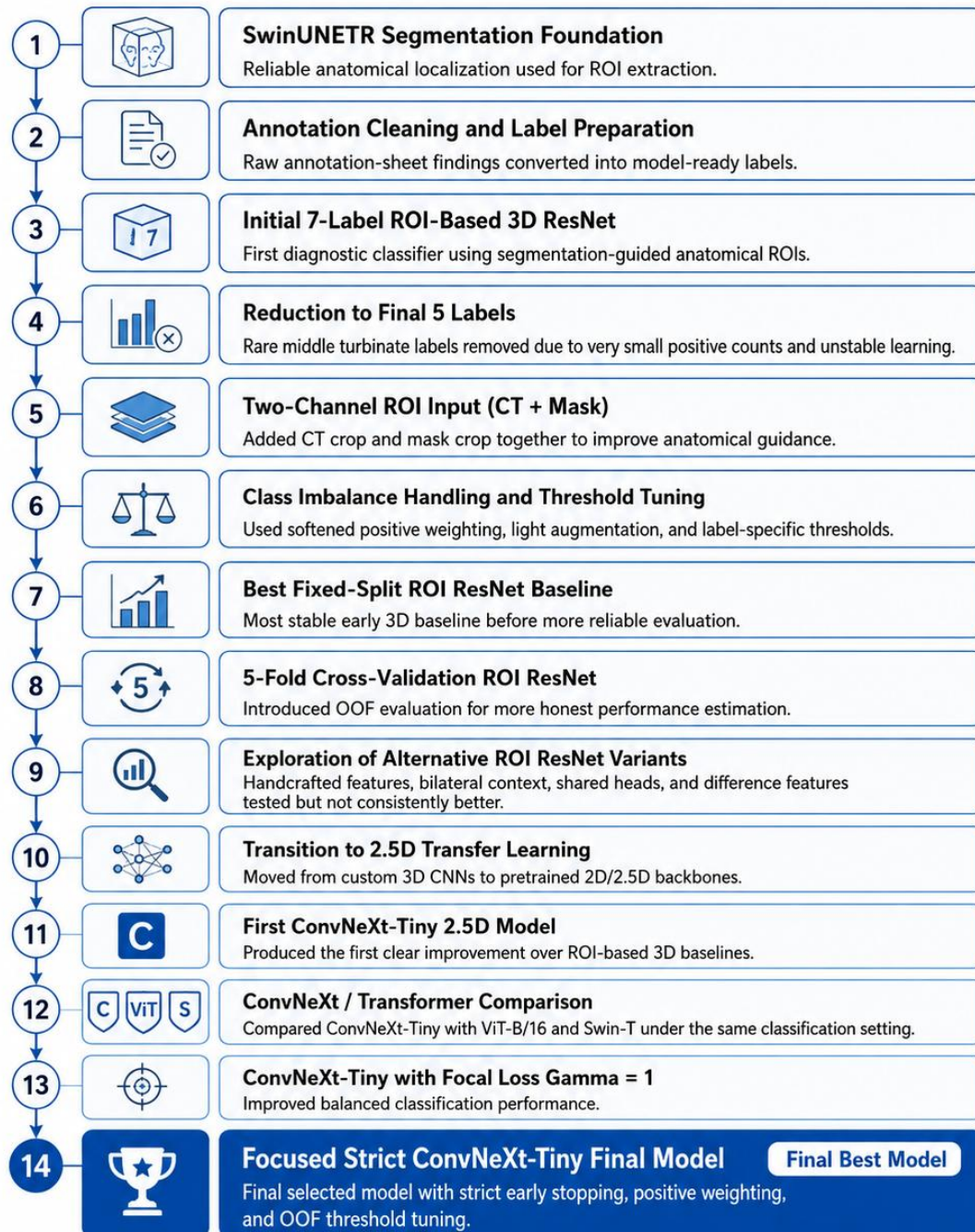
Additional diagnostic experiments were useful even when they were not selected. Handcrafted ROI statistics, tighter pseudo-septum crops, bilateral context, shared heads, signed and absolute difference features, and ground-truth mask extraction were all examined. These experiments showed that segmentation quality was helpful but not the only bottleneck; the main limitations were small sample size, rare positive labels, indirect septum and turbinate targets, and threshold instability.

The major improvement came from replacing the custom 3D encoder with a 2.5D transfer-learning classifier. Each 3D ROI was converted into axial, coronal, and sagittal views, enabling the use of pretrained image backbones. ConvNeXt-Tiny was the strongest direction because it improved feature extraction while remaining suitable for the limited dataset size.

In the later focused experiments, ConvNeXt-Tiny with focal loss gamma equal to 1, positive class weighting, light augmentation, strict early stopping, ReduceLROnPlateau, and threshold tuning achieved the strongest fold-level macro F1. Swin-T was retained as an important transformer comparison experiment, but it did not outperform the final ConvNeXt-Tiny configuration.

Overall, the experimental history shows that the final pipeline was shaped by systematic failure analysis as much as by successful results. The project moved from segmentation, to ROI-based 3D diagnosis, to honest 5-fold evaluation, and finally to 2.5D ConvNeXt-Tiny transfer learning because each stage exposed a specific limitation of the previous design.

Timeline of Major Classification Experiments



This timeline summarizes the major methodological stages of classification-model development, from early ROI-based 3D baselines to the final ConvNeXt-Tiny model.

Figure 14: Timeline of major classification experiments

Table 9: Timeline of major exploratory experiments and outcomes

Stage	Experiment / Idea	Main observation	Decision
1	SwinUNETR segmentation foundation	Mean Dice about 0.87; masks localized maxillary sinuses, nasal cavities, and nasopharynx.	Kept as anatomical localization stage.
2	Annotation cleaning and multi-label formulation	Workbook findings were converted into five independent abnormality labels; normality was represented by all-zero targets.	Adopted for all diagnostic experiments.
3	Initial seven-label ROI 3D ResNet	First full segmentation-guided diagnostic classifier; middle turbinate labels failed because of very small positive counts.	Reduced task to five labels.
4	Five-label 3D ROI classifier with CT + mask input	Two-channel ROI input and separate label heads gave the most stable 3D ResNet direction.	Kept as main 3D baseline.
5	Imbalance handling and augmentation	Softened positive weights, light CT augmentation, and label-specific thresholds reduced some instability.	Retained only stable components.
6	Best fixed-split 3D ROI ResNet	Approximate macro F1 around 0.63, but performance depended strongly on the selected split and thresholds.	Used as fixed-split baseline only.
7	Five-fold 3D ROI ResNet evaluation	OOF macro F1 about 0.588; showed that fixed-split results were optimistic.	Adopted 5-fold OOF evaluation.
8	Alternative ROI and head designs	Handcrafted features, tighter septum ROIs, bilateral context, shared heads, and difference features did not consistently improve results.	Rejected in favor of simpler stable design.
9	Ground-truth mask experiment	Oracle masks improved some ranking behavior but did not clearly improve thresholded OOF F1.	Confirmed labels/data were main bottleneck.
10	2.5D transfer-learning transition	Converted ROIs into axial, coronal, and sagittal views and used pretrained backbones.	Adopted for final classifiers.
11	ConvNeXt-Tiny focal-loss experiments	ConvNeXt-Tiny with focal loss $\gamma=1$ produced the strongest fold-level macro F1 among tested backbones.	Selected as final direction.
12	Swin-T transformer comparison	Swin-T was competitive and useful as a transformer baseline, but remained below ConvNeXt-Tiny.	Kept as comparison, not final model.

Macro F1 Comparison Across Individual Classification Experiments

Detailed comparison of individual experiment variants across the classification-development timeline.

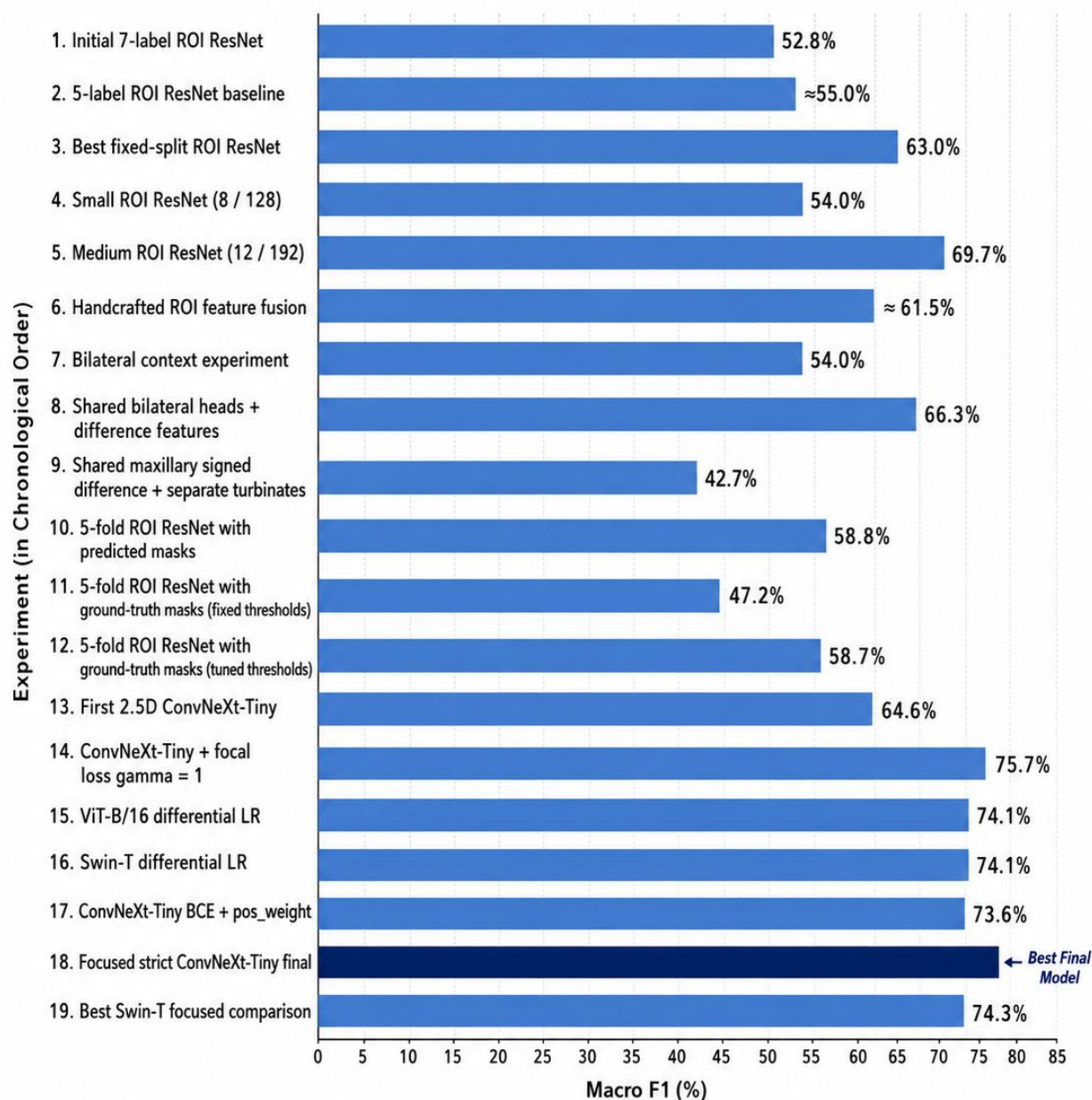


Figure 15: Macro F1 comparison across experiments.

4.11 Final Selected Classification Model

The final selected classification backbone is ConvNeXt-Tiny. It was selected after comparing it with Swin-T under the same segmentation-guided 2.5D ROI classification setting. Swin-T was retained as a transformer-based comparison model, while ConvNeXt-Tiny achieved stronger fold-level and out-of-fold performance in the final experiments.

The choice of ConvNeXt-Tiny was based on both experimental and methodological reasons. Experimentally, it achieved the strongest final classification metrics among the compared backbones. Methodologically, its convolutional design is suitable for localized ROI images, where local texture, boundary, and shape patterns are important. This made it a practical choice for limited-data medical-image classification.

4.11.1 Original ConvNeXt-Tiny Architecture

ConvNeXt-Tiny is a modern convolutional neural network introduced as part of the ConvNeXt family. It preserves the basic locality of convolutional neural networks while adopting several design ideas that became common in transformer-era vision models. The model is organized as a hierarchical feature extractor with a stem layer followed by four main feature stages.

The stem, sometimes described as a patchifying stem, converts the input image into an initial feature representation. Instead of processing the raw image directly at all later layers, the stem produces a lower-resolution feature map with more feature channels. This allows the model to begin extracting visual patterns while reducing spatial resolution.

ConvNeXt-Tiny contains four main stages with block depths of 3, 3, 9, and 3. The feature channel dimensions increase across these stages from 96 to 192, 384, and finally 768. This means that early stages preserve more spatial detail, while deeper stages learn richer and more abstract features. Between stages, downsampling reduces the spatial size of the feature maps while increasing the number of channels.

Inside a ConvNeXt block, a large 7×7 depthwise convolution is used to capture a wider local context efficiently. Depthwise convolution processes each feature channel separately in the spatial dimension, while later pointwise or MLP-style layers mix information between channels. Layer normalization helps stabilize feature values, and residual connections help preserve useful

information and improve training stability.

At the end of the original ConvNeXt-Tiny model, global average pooling summarizes the final feature maps into one feature vector. The original classification head then maps this vector to ImageNet classes. In this project, this original classification head was not used directly; it was replaced and adapted for sinus CT abnormality prediction.

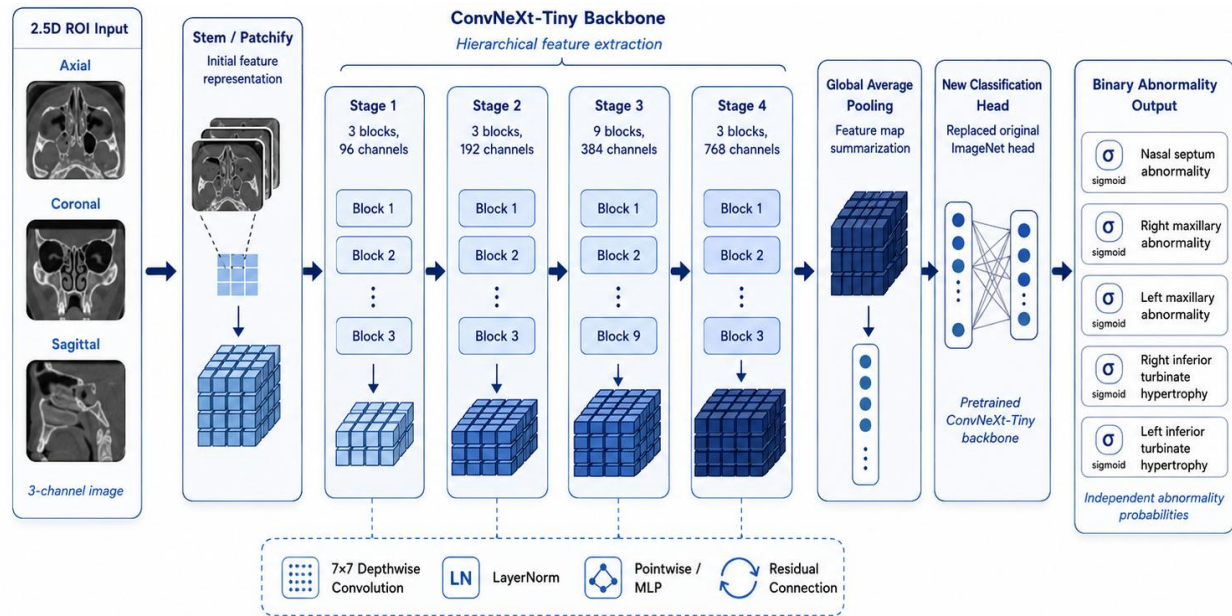


Figure 16: ConvNeXt-Tiny architecture adapted for 2.5D sinus ROI classification

4.11.2 Adaptation for 2.5D Sinus ROI Classification

The original ConvNeXt-Tiny model was designed for natural-image classification. In this project, it was adapted to classify sinus CT ROIs. Each input sample is a 2.5D ROI image constructed from three orthogonal CT views: axial, coronal, and sagittal. These views are stacked as three channels, similar in shape to an RGB image, but the channels represent medical planes rather than color channels.

The use of 2.5D ROI images provides a compromise between full 3D modeling and simple 2D slice classification. It preserves information from three anatomical planes while allowing the use of pretrained two-dimensional backbones. This is important because the dataset contains only 130 matched CT cases, making fully 3D classification more difficult and more prone to overfitting.

The original ImageNet classification head was replaced with a project-specific classification head. The final output is interpreted as independent binary abnormality probabilities, which are assembled into the five-label CT-case prediction vector. Sigmoid activation is used because each abnormality target is treated as a presence-or-absence decision rather than as a mutually exclusive class.

Table 10: Final ConvNeXt-Tiny classification training configuration

Parameter	Value / Setting
Backbone	ConvNeXt-Tiny
Input representation	2.5D ROI image
Input channels	3 channels: axial, coronal, sagittal
Pretraining	ImageNet pretrained backbone
Output formulation	Independent binary abnormality probabilities
Final targets	5 selected sinus CT abnormality labels
Activation	Sigmoid
Loss function	Focal loss
Focal gamma	1
Imbalance handling	Positive class weighting
Augmentation	Light on-the-fly augmentation
Scheduler	ReduceLROnPlateau
Optimizer	AdamW
Learning Rate Backbone	1e-4
Learning Rate Head	1e-4
Overfitting control	Strict early stopping
Evaluation	Balanced 5-fold out-of-fold evaluation
Thresholding	Label-specific OOF threshold tuning

4.11.3 Loss Function, Thresholding, and OOF Evaluation

The final classification experiments used focal loss with gamma equal to 1 and positive class weighting. Focal loss was selected because the classification labels are imbalanced and because some samples are more difficult than others. Positive class weighting was used to reduce the

tendency of the model to favor negative predictions for rare labels.

A fixed threshold of 0.5 was not used as the only final decision rule because the five labels have different prevalence and probability distributions. Instead, threshold tuning was performed on out-of-fold predictions to select label-specific probability cutoffs. This allowed each abnormality label to use a threshold better matched to its validation behavior.

The final evaluation used balanced 5-fold out-of-fold prediction. In this setup, each CT case is predicted by a model that did not train on it. The predictions from all validation folds are then combined to compute OOF macro F1 and per-label metrics. This evaluation is more reliable than a single split for this project because the dataset is small and imbalanced.

4.12 Implementation Environment

The experiments were implemented in a notebook-based Python environment. The segmentation pipeline used PyTorch and MONAI for 3D medical image processing, transforms, training, and sliding-window inference. The classification pipeline used PyTorch, TorchVision, pandas, NumPy, scikit-learn, and Matplotlib. Training was performed in a Kaggle GPU environment. The final notebooks produced checkpoint files, ROI caches, OOF prediction tables, threshold tables, metrics CSV files, and plots used for analysis.

4.13 Chapter Summary and Research Contribution

The methodology combines 3D anatomical segmentation with ROI-based 2.5D multi-label classification. The major methodological decision is to avoid using the full CT volume as an unlocalized classifier input and instead use segmentation-guided anatomical crops. This makes the pipeline more interpretable and allows each output label to be associated with a clinically meaningful region. The final selected classifier is ConvNeXt-Tiny, while Swin-T is documented as a previous transformer comparison experiment rather than a second final model.

Chapter Five: Research Results

5.1 Introduction

This chapter presents the experimental results of the proposed sinus CT analysis system. The results are reported in two main parts: segmentation results and classification results. Segmentation results evaluate the SwinUNETR model as the anatomical localization stage. Classification results evaluate the final multi-label abnormality prediction stage and compare ConvNeXt-Tiny with Swin-T.

5.2 Tools and Evaluation Methods

The experiments were implemented using Python, PyTorch, MONAI, Torchvision, NumPy, pandas, scikit-learn, and Matplotlib. Training was performed in a Kaggle GPU environment. The segmentation model was evaluated with Dice, IoU, precision, recall, inference time, and throughput. The classification models were evaluated with macro accuracy, macro precision, macro recall, macro specificity, macro balanced accuracy, macro F1, macro AUC, macro average precision, overall label accuracy, and exact match accuracy.

5.2.1 Experimental Environment

The segmentation and classification experiments were developed as notebook-based workflows. This allowed step-by-step inspection of data loading, preprocessing, ROI extraction, model training, metric computation, and figure generation. The final artifacts included trained checkpoints, ROI cache files, fold summaries, out-of-fold predictions, tuned thresholds, per-label metrics, and plots.

5.2.2 Evaluation Metrics Used in the Final Experiments

Segmentation was evaluated using mean Dice, mean IoU, precision, recall, inference time, and throughput. Classification was evaluated using label-wise and macro metrics. Because the task is multi-label and imbalanced, macro F1 and out-of-fold metrics were emphasized over simple accuracy. Exact match accuracy was also reported, but it is naturally strict because it requires all five labels of each case to be correct at the same time.

5.3 Segmentation Results

The SwinUNETR model achieved strong segmentation performance on the test set. The mean Dice score was 0.8743 and the mean IoU was 0.8102. The high recall value indicates that the model generally captured most of the reference anatomical regions. The segmentation results are suitable for the intended purpose of guiding ROI extraction in the classification stage.

5.3.1 Quantitative Segmentation Results



Figure 17: SwinUNETR training and validation curves

Table 11: Overall segmentation performance on the test set

Metric	Value
Mean Dice	0.8743
Mean IoU	0.8102
Mean Precision	0.8424
Mean Recall	0.9158
Inference time	0.34 s / volume
Throughput	2.90 volumes / s

The segmentation results indicate that the SwinUNETR model was able to provide reliable anatomical localization for the downstream classification stage. The mean Dice score of 0.8743 and mean IoU of 0.8102 show strong overlap between the predicted masks and the reference anatomical labels. The recall value of 0.9158 is particularly important for this project because the segmentation masks are used to extract regions of interest. A high recall means that the model

usually captures most of the target anatomical region, reducing the risk that relevant CT information is excluded before classification. Although precision is slightly lower than recall, this is acceptable for the purpose of ROI generation because a slightly larger anatomical crop is less harmful than missing part of the target region.

Table 12: Per-class segmentation performance

Class	Dice	IoU	Precision	Recall
Background	0.9896	0.9796	0.9946	0.9846
Right maxillary sinus	0.9107	0.8543	0.8797	0.9476
Left maxillary sinus	0.8697	0.8104	0.8503	0.9053
Right nasal cavity	0.8009	0.7010	0.7440	0.8729
Left nasal cavity	0.8079	0.7138	0.7520	0.8772
Nasopharynx	0.8669	0.8024	0.8338	0.9075

The per-class results show that the maxillary sinuses achieved the strongest non-background segmentation performance, especially the right maxillary sinus with a Dice score of 0.9107. This is useful for the classification stage because right and left maxillary abnormality labels depend directly on the corresponding maxillary sinus ROIs. The nasal cavity classes achieved lower Dice scores than the maxillary sinuses, with Dice values of 0.8009 and 0.8079 for the right and left nasal cavities. This is expected because the nasal cavity has a more complex and irregular shape than the maxillary sinus and contains narrow anatomical spaces. However, these results remained sufficient for contextual ROI extraction, especially for nasal septum and inferior turbinate labels, where the nasal cavity masks are used as anatomical guides rather than final diagnostic outputs.

5.3.2 Qualitative Segmentation Results

In addition to quantitative metrics, qualitative inspection was performed to verify that the predicted masks followed the expected sinonasal anatomy. This step is important because high global segmentation scores may hide local errors that affect ROI extraction. The qualitative examples were inspected in axial, coronal, and sagittal views to ensure that the predicted regions corresponded to the right maxillary sinus, left maxillary sinus, right nasal cavity, left nasal cavity, and nasopharynx.

Figure 18 shows a representative segmentation result on a test case. The predicted mask generally follows the anatomical boundaries of the sinus and nasal cavity regions. The maxillary sinus regions are clearly localized, which supports their use as direct ROIs for right and left maxillary abnormality classification. The nasal cavity regions also provide useful contextual localization for nasal septum and inferior turbinate-related classification, even though the septum and turbinates are not directly annotated as segmentation classes in the NasalSeg dataset.

Some segmentation difficulty can still occur around narrow nasal passages, thin bony boundaries, and irregular mucosal regions. These areas are more challenging because small voxel-level shifts may reduce Dice or IoU even when the predicted region remains clinically useful for ROI extraction. Therefore, the segmentation output should be interpreted as an anatomical localization tool rather than a final clinical segmentation system. For the purpose of this project, the qualitative results confirm that SwinUNETR provides sufficiently stable anatomical guidance for the downstream ROI-based classification stage.

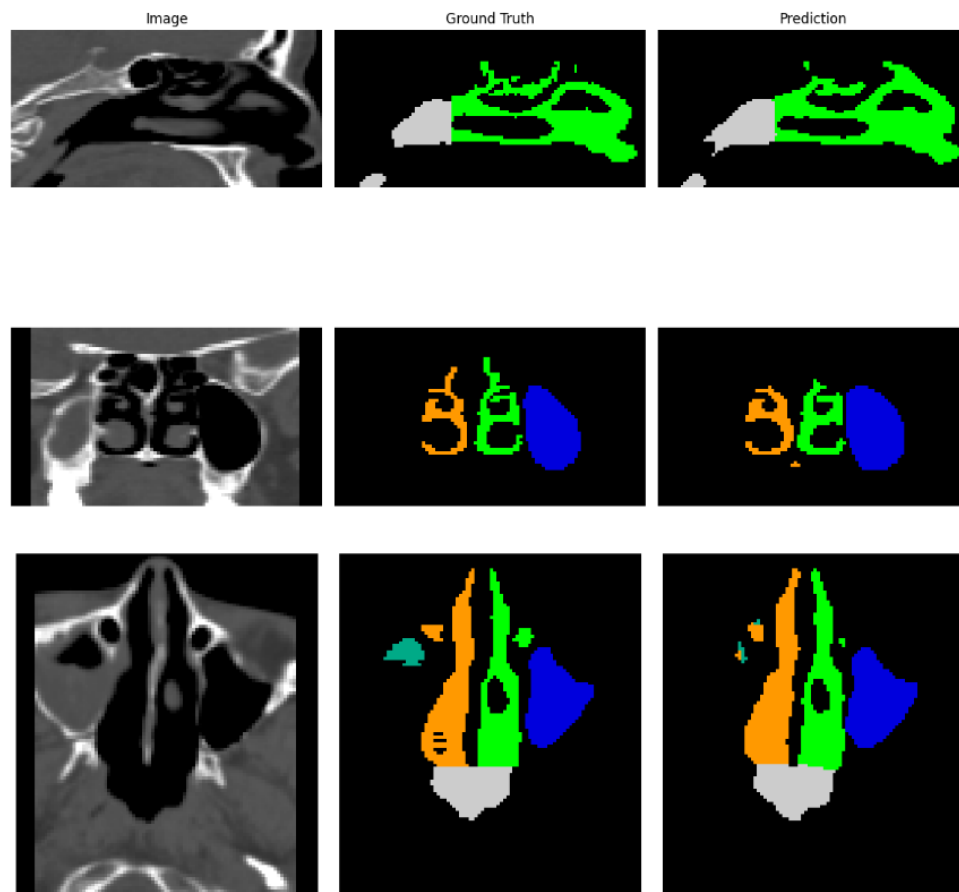


Figure 18: Example segmentation prediction on a test case

5.4 Classification Results

The classification stage evaluates the ability of the proposed system to predict five abnormality labels from segmentation-guided 2.5D ROI inputs. The task is evaluated as a multi-label problem at the CT-case level because more than one abnormality may appear in the same case. Therefore, the classification results are not interpreted using a single accuracy value only. Instead, the final analysis emphasizes out-of-fold macro F1, per-label F1, label-wise confusion statistics, and threshold-tuned predictions.

ConvNeXt-Tiny was selected as the final classifier after comparison with Swin-T under the same final evaluation setting. The analysis below first describes the training behavior, then reports the final OOF metrics, then analyzes each label separately. This organization is used because the dataset is small and imbalanced, and because some labels are more difficult than others.

5.4.1 Training Behavior and Overfitting Analysis

The training curves were used to monitor learning behavior across the five folds. The training loss generally decreased across epochs, which indicates that the model was able to learn patterns from the ROI-based training data. However, the validation loss was more unstable and increased in some folds after the early epochs.

This behavior is expected in this project because each validation fold contains only about one fifth of the 130 CT cases. Therefore, each validation fold contains a small number of positive examples, especially for the right and left maxillary abnormality labels. A few confident wrong predictions can strongly affect validation loss. This effect is amplified by focal loss and positive class weighting, because difficult positive examples receive more training importance.

For this reason, the loss curves were not used alone to judge the final classifier. They were used mainly to observe overfitting and support early stopping. The final model selection relied more on out-of-fold macro F1, per-label F1, and threshold-tuned metrics because these metrics better reflect the multi-label and imbalanced nature of the task.

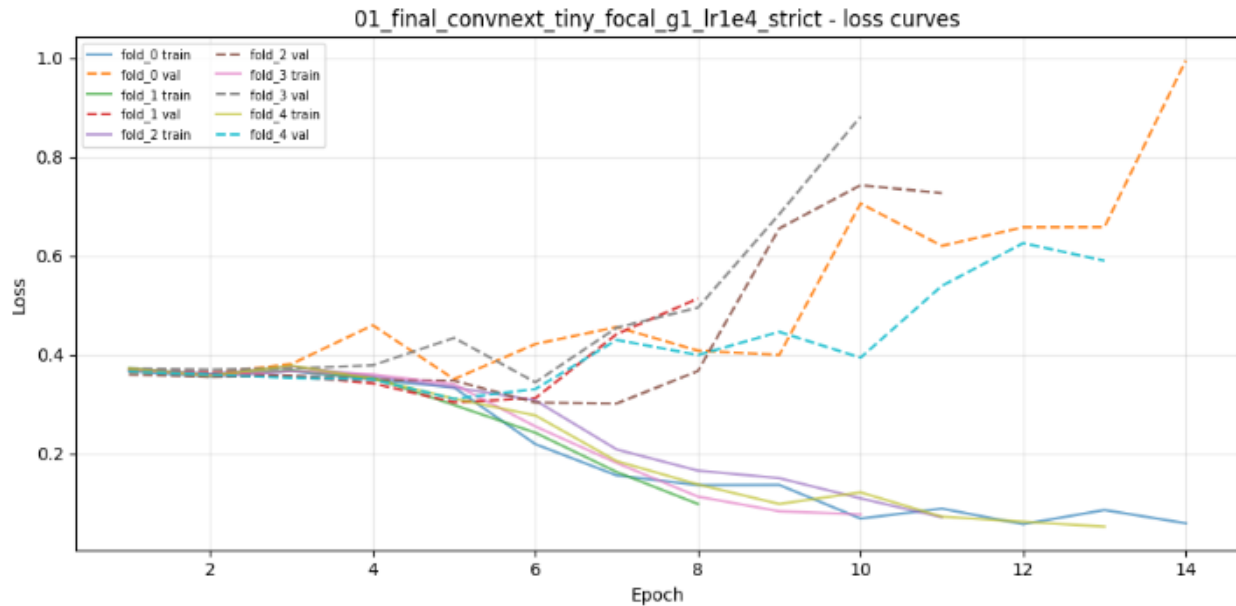


Figure 19: ConvNeXt-Tiny loss curves

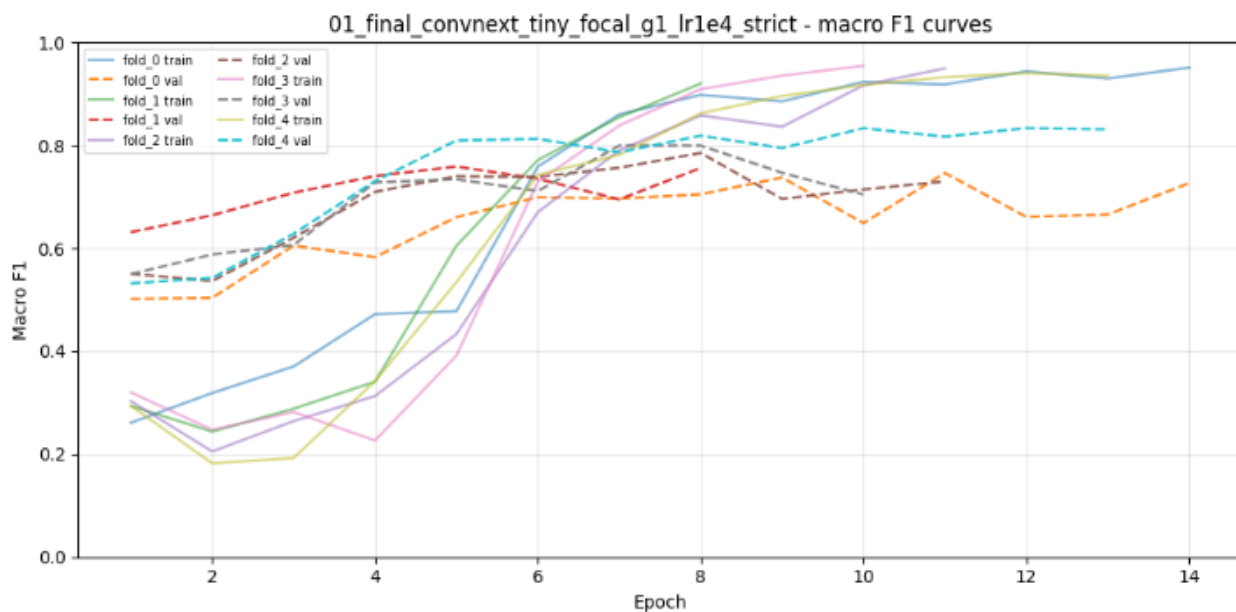


Figure 20: ConvNeXt-Tiny macro F1 curves

5.4.2 Final ConvNeXt-Tiny Classification Metrics

The final ConvNeXt-Tiny classifier achieved a fold-level macro F1 of 0.7849 and an out-of-fold macro F1 of 0.6928. The OOF macro F1 is treated as the main final classification metric because each CT case is evaluated by a model that did not train on it. This is more reliable than depending only on a single train-test split, especially with a limited dataset.

The model also achieved an overall label accuracy of 0.7877. This indicates that most individual

binary label decisions were correct. However, exact match accuracy was lower at 0.3846 because it requires all five labels of a case to be predicted correctly at the same time. In multi-label classification, this metric is naturally strict: one wrong label makes the whole case incorrect under exact match evaluation.

Table 13: Final ConvNeXt-Tiny classification summary

Metric	Value	Interpretation
Fold-level macro F1	0.7849	Average validation macro F1 across folds
OOF macro F1	0.6928	Main final metric computed from combined out-of-fold predictions
Overall label accuracy	0.7877	Correctness across all individual label decisions
Exact match accuracy	0.3846	Strict case-level metric requiring all five labels to be correct
Best final classifier	ConvNeXt-Tiny	Selected over Swin-T in the final comparison

5.4.3 Per-Label Performance Analysis

The per-label metrics are more informative than a single aggregate score because each abnormality has a different positive count and anatomical difficulty. Therefore, Table 14 reports threshold, accuracy, precision, recall, and F1 for each final label using the OOF predictions.

Table 14: OOF per-label metrics for ConvNeXt-Tiny

Label	Threshold	Accuracy	Precision	Recall	F1
left_inferior_turbinate_abnormal	0.49	0.7308	0.7049	0.7167	0.7107
left_maxillary_abnormal	0.42	0.8462	0.5600	0.6087	0.5833
nasal_septum_abnormal	0.44	0.6846	0.6786	0.8028	0.7355
right_inferior_turbinate_abnormal	0.46	0.7769	0.7385	0.8000	0.7680
right_maxillary_abnormal	0.43	0.9000	0.7647	0.5909	0.6667

5.4.4 Per-Label Confusion Analysis

Because the task is multi-label, each abnormality label can be evaluated as a separate binary classification problem. Therefore, per-label true positives, false positives, true negatives, and false negatives provide a clearer interpretation than a single dominant-label confusion matrix. This analysis shows where the model tends to miss positive cases or produce false alarms for each abnormality.

Table 15: Per-label binary confusion statistics for ConvNeXt-Tiny OOF predictions

Label	TP	FP	TN	FN	Main observation
left_inferior_turbinate_abnormal	43	18	52	17	Balanced but still misses some positives
left_maxillary_abnormal	14	11	96	9	Limited by low positive count
nasal_septum_abnormal	57	27	32	14	High recall but more false positives
right_inferior_turbinate_abnormal	48	17	53	12	Strongest positive detection
right_maxillary_abnormal	13	4	104	9	High precision but lower recall

5.4.5 Dominant-Label Confusion Matrix

The dominant-label confusion matrix provides a compact visualization of the strongest predicted abnormality per case. However, it should not be interpreted as the main classification metric because the task is multi-label. A CT case may contain more than one abnormality, while the dominant-label matrix compresses each case into one true label and one predicted label.

This compression can hide partially correct predictions. For example, a case with both septum abnormality and maxillary abnormality may be shown as confused if the dominant confidence differs, even when one or more labels were correctly predicted. Therefore, Figure 21 is used only as a supporting visualization. The main evaluation remains the OOF macro F1, per-label F1, and per-label binary confusion statistics.

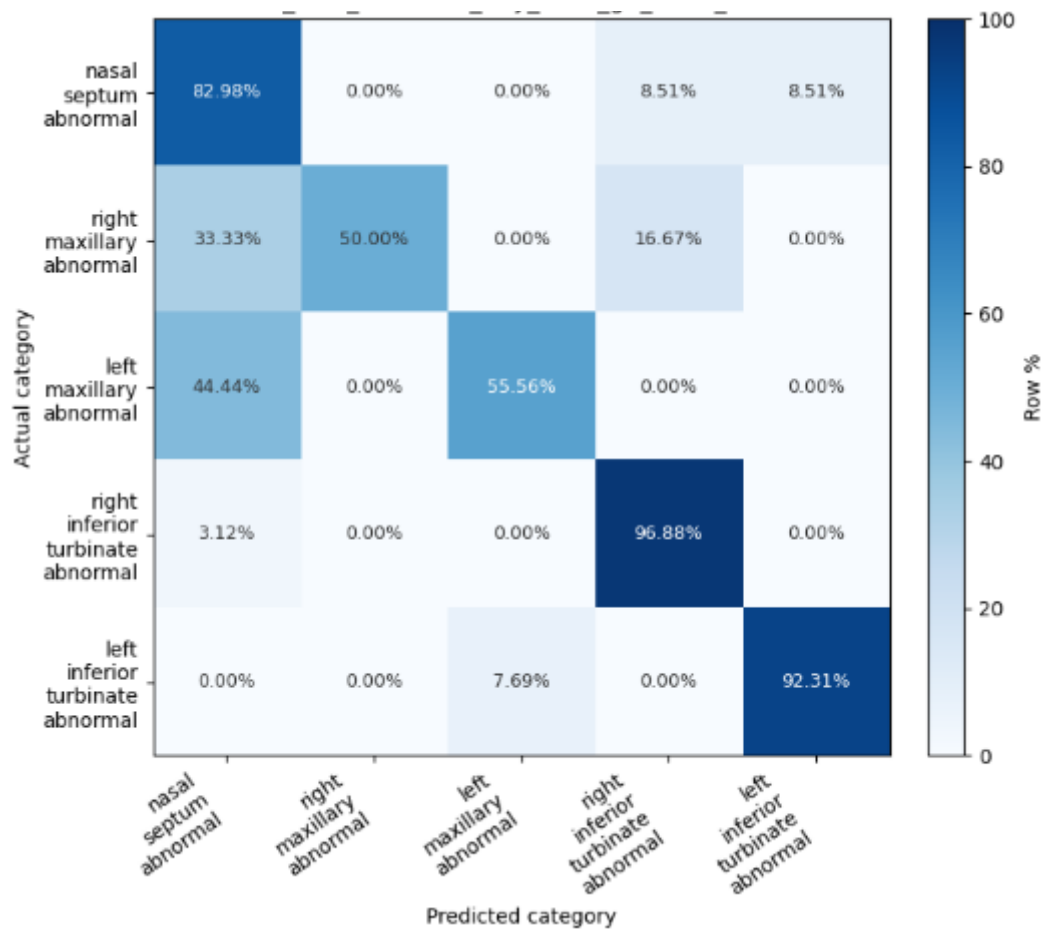


Figure 21: ConvNeXt-Tiny 5×5 dominant-label confusion matrix

5.4.6 ROC Analysis

ROC curves were used as supporting analysis to evaluate how well the model ranks positive cases above negative cases across different thresholds. However, ROC curves can appear optimistic when the negative class is much larger than the positive class. For this reason, ROC analysis should not replace per-label F1 or average precision in this project. Therefore, ROC is interpreted as supporting threshold-dependent analyses, while the final model selection remains based mainly on OOF macro F1 and per-label performance.

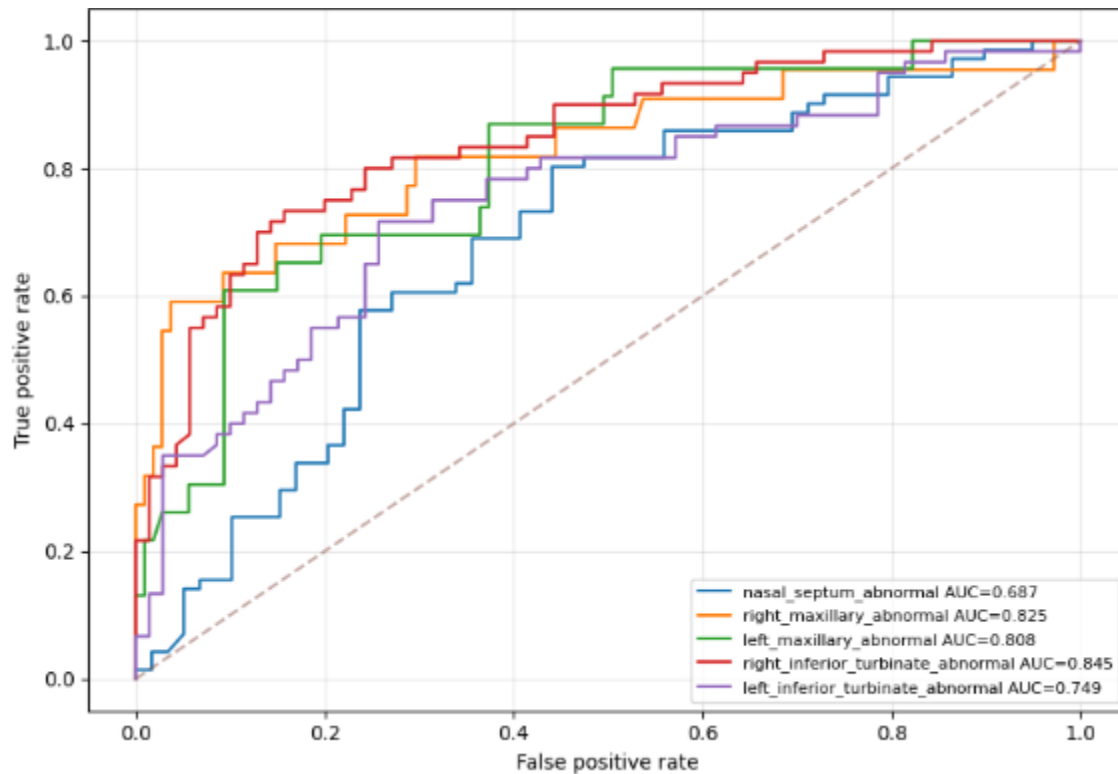


Figure 22: ROC curves for the final ConvNeXt-Tiny model

5.4.7 Internal Model Comparison

The final ConvNeXt-Tiny model was not selected from a single experiment. Earlier ROI-based 3D ResNet experiments showed that segmentation-guided classification was feasible, but performance was sensitive to data split and thresholds. When evaluated with 5-fold OOF prediction, the 3D ROI ResNet baseline achieved an OOF macro F1 of approximately 0.588, confirming that the earlier fixed-split result was optimistic.

The transition to 2.5D transfer learning improved the classification stage by allowing the use of pretrained image-classification backbones. In the final comparison, ConvNeXt-Tiny outperformed Swin-T in fold-level macro F1, OOF macro F1, overall label accuracy, and exact match accuracy. Therefore, ConvNeXt-Tiny was selected as the final classification backbone.

Table 16: Internal classification model comparison

Model / stage	Evaluation setting	Main result	Decision
Fixed-split 3D ROI ResNet	Single split	Macro F1 $\approx$ 0.63	Used only as fixed-split baseline

5-fold 3D ROI ResNet	OOF evaluation	OOF macro F1 $\approx$ 0.588	Showed fixed split was optimistic
Swin-T	5-fold OOF	OOF macro F1 0.6565	Kept as transformer comparison
ConvNeXt-Tiny	5-fold OOF	OOF macro F1 0.6928	Selected as final classifier

5.5 Comparison with Previous Works

Comparison with previous works must be interpreted carefully because the proposed project does not solve exactly the same task as most related studies. Many previous works focus on segmentation only, while others focus on binary maxillary sinus anomaly classification. In contrast, this project combines anatomical segmentation with five-label multi-label abnormality prediction. Therefore, the comparison is not a direct competition on the same dataset and task, but an analysis of how the proposed system is positioned relative to existing work.

For segmentation, the proposed SwinUNETR model achieved a mean Dice score of 0.8743 and mean IoU of 0.8102 on the test set. This result is consistent with the literature showing that 3D deep-learning models are effective for sinus and paranasal anatomical segmentation. Compared with segmentation-only studies, the main difference is that segmentation in this project is not the final output. Instead, it is used as an anatomical guide for ROI extraction before classification.

For classification, the proposed ConvNeXt-Tiny model achieved an out-of-fold macro F1 score of 0.6928 for five abnormality labels. This task is more complex than binary maxillary anomaly classification because each CT case may contain multiple simultaneous abnormalities. In addition, some target labels do not have direct segmentation masks, especially nasal septum abnormality and inferior turbinate hypertrophy. For this reason, the classification result should be understood as a multi-label research prototype rather than a direct replacement for narrower classification studies.

Table 17: Comparison of the proposed model with previous works

Work	Task	Data / scope	Main result	Relation to this project
Zhang et al., NasalSeg [1]	Sinonasal anatomical segmentation	130 CT scans, five structures	Public dataset and benchmark for nasal/paranasal segmentation	Provides the anatomical basis used for ROI extraction

Whangbo et al. [6]	Paranasal sinus segmentation	Sinusitis CT data	Best F1 84.29% on normal test set and 79.32% on abnormal test set	Related segmentation task, but no abnormality classification
Song et al. [10]	Comparison of segmentation architectures	Paranasal sinus CT	Swin UNETR Dice 0.830, Jaccard 0.719	Supports using SwinUNETR-style segmentation for sinus CT
Bhattacharya et al. [7]	Maxillary anomaly classification	Maxillary sinus anomaly task	AUPRC 0.79 with 10% annotated data	Related classification work, but narrower than five-label prediction
Bhattacharya et al. [8]	Maxillary anomaly classification	Maxillary sinus anomaly task	AUROC 0.85 using supervised contrastive learning	Shows value of deep representation learning, but not multi-label sinus CT
Proposed segmentation stage	Anatomical localization	NasalSeg CT cases	Mean Dice 0.8743, mean IoU 0.8102	Provides ROI guidance for classification
Proposed classification stage	Five-label sinus CT abnormality prediction	130 matched CT cases	ConvNeXt-Tiny OOF macro F1 0.6928	Combines segmentation-guided ROIs with multi-label classification

The comparison shows that the proposed system is different from most previous work because it combines two stages: anatomical localization and abnormality prediction. Its segmentation performance is strong enough to support ROI extraction, while the classification results show promising but data-limited multi-label prediction. The main advantage of the proposed system is anatomical interpretability. The main limitation is that the classification stage is trained on only 130 cases with strong label imbalance, especially for the maxillary abnormality labels.

5.6 Discussion of Results

The results show that the proposed system achieved its main goal as a research prototype: combining anatomical segmentation with ROI-based abnormality classification. The segmentation stage provided reliable anatomical localization, while the classification stage showed that selected sinus CT abnormalities can be predicted from anatomy-guided 2.5D ROI inputs. However, the results also show that classification performance remains limited by dataset

size, label imbalance, and the absence of direct masks for some anatomical targets.

5.6.1 Segmentation-Guided ROI Extraction

Segmentation-guided ROI extraction was one of the most important design decisions in the project. Instead of asking the classifier to analyze the full CT volume without localization, the system first identifies relevant sinonasal anatomical structures and then uses them to guide the classification inputs. This design is medically appropriate because the target abnormalities are region-dependent.

The segmentation results were strong enough for this role. The high recall is especially useful because ROI extraction depends on capturing the target anatomical region. Missing part of the relevant anatomy could remove useful information before classification, while a slightly larger ROI is usually less harmful. Therefore, the segmentation stage should be interpreted as an anatomical localization tool that supports classification, not as a standalone clinical segmentation product.

5.6.2 Classification Performance and Label Difficulty

The classification results show that ConvNeXt-Tiny was the strongest final classifier among the evaluated models. Its out-of-fold macro F1 of 0.6928 indicates that the model learned useful abnormality-related patterns from the segmentation-guided ROI inputs. The result is promising for a limited 130-case dataset, but it should be interpreted as research performance rather than clinical readiness.

The per-label results show that the model did not perform equally across all abnormalities. Septum and inferior turbinate-related labels were more stable, while right and left maxillary abnormality labels were weaker. This difference reflects the unequal difficulty of the labels. The turbinate and septum labels had more positive cases, while the maxillary labels had only 22 and 23 positives. Therefore, the lower maxillary performance is consistent with the limited positive sample count and should not be interpreted as a failure of the ROI-based design itself.

5.6.3 Effect of Limited Data and Imbalance

The main limitation affecting classification performance was the small and imbalanced dataset.

Focal loss, positive class weighting, threshold tuning, and 5-fold out-of-fold evaluation helped make the training and evaluation more appropriate for this setting. However, these techniques cannot fully compensate for the lack of positive examples.

This was most visible in the maxillary abnormality labels. With only 22 right maxillary and 23 left maxillary positive cases, the model had limited examples from which to learn stable abnormality patterns. In addition, abnormal findings may co-occur in the same CT case, which makes label-specific interpretation more difficult. These factors explain why macro F1 and per-label F1 are more meaningful than simple accuracy in this project.

5.6.4 Achievement of Research Objectives

The segmentation objective was strongly achieved. SwinUNETR localized the available NasalSeg anatomical structures with a mean Dice score of 0.8743 and mean IoU of 0.8102, which was sufficient for downstream ROI extraction.

The classification objective was achieved as a research prototype. ConvNeXt-Tiny achieved the best final classification performance among the evaluated backbones and produced meaningful out-of-fold predictions for the five selected abnormality labels. However, the classification stage remains limited by small dataset size, class imbalance, and indirect ROI definitions for septum and turbinate-related targets.

The comparison objective was also achieved. The project compared earlier ROI-based 3D classification experiments, Swin-T, and ConvNeXt-Tiny under increasingly reliable evaluation settings. The final selection of ConvNeXt-Tiny was supported by its stronger performance in the final 5-fold out-of-fold evaluation.

Overall, the project demonstrates a complete segmentation-assisted sinus CT analysis pipeline. Its main achievement is not only the final classifier, but the full workflow connecting anatomical localization, ROI extraction, 2.5D classification, and multi-label evaluation.

5.7 Chapter Summary

The results show that SwinUNETR provides strong anatomical segmentation and that ConvNeXt-Tiny is the best final classifier among the evaluated models. The main remaining weaknesses are related to label imbalance, multi-label co-occurrence, and lack of direct masks

for septum and turbinates. These issues guide the future work described in the final chapter.

Chapter Six: Conclusion

6.1 Introduction

This chapter summarizes the main conclusions of the project, identifies the most important limitations, and outlines future work needed to improve the proposed sinus CT analysis system. The project should be interpreted as a research prototype for segmentation-guided medical image analysis, not as a clinically deployable diagnostic system.

6.2 Conclusions

This project developed a segmentation-assisted AI system for sinus CT analysis. The proposed pipeline first uses SwinUNETR to segment key sinonasal anatomical structures, then extracts anatomy-guided regions of interest, and finally applies ROI-based classification to predict five selected abnormality labels: nasal septum abnormality, right maxillary abnormality, left maxillary abnormality, right inferior turbinate hypertrophy, and left inferior turbinate hypertrophy.

The segmentation objective was strongly achieved. SwinUNETR achieved a mean Dice score of 0.8743 and a mean IoU of 0.8102 on the test set, showing that the available NasalSeg structures can be localized with sufficient quality to support ROI extraction.

The classification objective was achieved as a research prototype. ConvNeXt-Tiny was selected as the final classifier because it achieved the best performance among the final compared models, with an out-of-fold macro F1 score of 0.6928. This result shows that segmentation-guided 2.5D ROIs can support abnormality prediction, while also revealing limitations caused by small dataset size, class imbalance, label co-occurrence, and missing direct masks for some anatomical targets.

The study also confirms that multi-label sinus CT evaluation must be interpreted carefully. Since a single CT case may contain several co-existing findings, per-label metrics and out-of-fold macro F1 are more suitable than relying only on a dominant-label confusion matrix. Overall, the project demonstrates a complete medical-AI workflow that connects anatomical segmentation, ROI extraction, and multi-label abnormality prediction.

6.3 Research Limitations

The project has several limitations that should be considered when interpreting the results:

- Limited dataset size: The final experiments used 130 matched CT cases, which is small for deep learning and limits generalization.
- Class imbalance: The right and left maxillary abnormality labels had only 22 and 23 positive cases, making these labels less stable than septum and turbinate labels.
- Missing direct anatomical masks: NasalSeg does not directly provide nasal septum or turbinate masks. Therefore, septum and inferior turbinate predictions relied on contextual nasal cavity ROIs rather than direct target masks.
- Binary label simplification: The annotation workbook was converted into binary abnormality labels. Therefore, the final classifier predicts presence or absence only and does not model severity, subtype, direction of septal deviation, or detailed radiological grading.
- Multi-label co-occurrence: Some abnormalities appeared together in the same CT case, making label-specific interpretation more difficult and limiting the usefulness of dominant-label visualization.
- No external validation: The model was not tested on CT data from another hospital, scanner, or acquisition protocol.
- Research-prototype status: The system has not been prospectively tested or validated in a real clinical workflow and should not be considered clinically deployable.

6.4 Ethical and Clinical Deployment Considerations

Because this project is related to medical imaging, its results must be interpreted within a clinical safety context. The proposed system is intended only as a research prototype and should not be used as an autonomous diagnostic tool. Any future clinical use would require external validation, radiologist review, and prospective testing before deployment.

The system should be considered a decision-support or educational tool rather than a

replacement for radiologists or ENT physicians. The final clinical decision must remain under expert medical supervision, especially because the classification stage is affected by limited data, class imbalance, and missing direct masks for some anatomical targets.

Data privacy is also important in future development. CT volumes and clinical annotations should be anonymized before model training or evaluation. Any future dataset expansion should follow institutional rules for patient privacy, data access, and ethical approval.

Future clinical deployment would also require uncertainty estimation and calibration. The model should not only output abnormality probabilities, but also provide a reliable indication of confidence and possible failure cases. This is especially important for rare labels and anatomically indirect targets. Therefore, the current system should be treated as a foundation for further research rather than a validated clinical product.

6.5 Challenges and Future Work

The main future direction is dataset expansion. More CT cases are needed, especially cases with right and left maxillary abnormalities. Increasing the number of positive maxillary cases would help the classifier learn more stable visual patterns and reduce the effect of label imbalance.

A second important direction is improving anatomical annotation. The current dataset provides masks for maxillary sinuses, nasal cavities, and nasopharynx, but it does not provide direct masks for the nasal septum or inferior turbinates. Adding expert segmentations for these structures would allow more precise ROI extraction and reduce dependence on contextual proxy regions.

Future work should also improve the diagnostic labels. The current model predicts binary presence or absence for the five selected abnormalities. A larger radiologist-reviewed dataset could support more detailed labels, such as severity, subtype, or direction of septal deviation. This would make the system more clinically informative, but it requires enough positive cases for each category.

External validation is another necessary step. The model should be evaluated on CT scans from another hospital, scanner, or acquisition protocol to determine whether it generalizes beyond the current dataset. Without external validation, the reported results should be understood as internal

research performance only.

From a modeling perspective, larger datasets may allow stronger hybrid 2.5D/3D classifiers. The current ConvNeXt-Tiny model was suitable for limited data, but future work can compare it with volumetric models, multi-ROI fusion networks, or architectures that explicitly combine anatomical context across regions.

Finally, a future clinical interface could display CT slices, segmentation masks, extracted ROIs, predicted labels, and prediction confidence together. Such an interface would be useful for educational or decision-support research, while keeping the final interpretation under expert medical supervision.

Table 18: Summary of main findings, limitations, and future work directions

Finding / limitation	Implication	Future direction
SwinUNETR achieved strong segmentation performance	Segmentation can guide ROI extraction	Improve masks and add direct annotations for missing structures
ConvNeXt-Tiny outperformed Swin-T	Modern CNN backbone was more stable for this dataset	Use ConvNeXt-Tiny as the final baseline for future comparisons
Maxillary labels were less stable	Low positive counts affected learning and recall	Collect more maxillary-positive CT cases
Septum and turbinate masks were unavailable	Contextual ROIs were necessary approximations	Add direct septum and turbinate segmentation masks
Binary labels simplified detailed annotations	Severity and subtype were not modeled	Add radiologist-reviewed severity and subtype labels
No external validation was performed	Generalization to other hospitals/scanners is unknown	Test on external CT datasets
System is not clinically deployable yet	Clinical use requires safety validation	Add uncertainty estimation, calibration, and prospective review

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